

Amino acid metabolism and Metabolic roles of Folic acid and vitamin B12

Other sources of questions to use as study aids

Work the questions in Lippincott's

Work the questions in the BRS

Look at the questions in the PreTest book

Work the relevant questions in Kaplan

Work on these questions that we provide for you

Make up your own questions and test each other

1. A 10-year-old boy is brought to the physician by his parents because of a 3-week history of burning sensation during urination. Urinalysis shows presence of red blood cells and hexagonal crystals. Which of the following is an additional finding in this patient?
 - A. Decreased tubular reabsorption of tryptophan
 - B. Deficiency of cystathionine β -synthase
 - C. Increased circulating levels of homocysteine
 - D. Increased urinary excretion of cystine
 - E. Elevated levels of homogentisic acid in urine
2. A 1-month-old boy is brought to the physician by his parents for a follow-up examination. Newborn screening studies showed elevated phenylalanine levels in blood and urine. He was prescribed a phenylalanine restricted diet. Physical examination shows a delay in the developmental milestones. He is prescribed sapropterin (biopterin analog). Which of the following reactions most likely requires this coenzyme?
 - A. Conversion of DOPA to dopamine
 - B. Conversion of Tyrosine to DOPA
 - C. Conversion of 5-hydroxy tryptophan to serotonin
 - D. Melanin synthesis
 - E. Conversion of branched chain amino acids to branched chain keto acids
3. A research group is studying the effects of a diet in a population that subsists on a protein source that is deficient in methionine. Synthesis of which of the following amino acids is most likely impaired in these subjects?
 - A. Alanine

- B. Tyrosine
- C. Cysteine
- D. Phenylalanine
- E. Glycine

4. A 55-year-old woman is brought to the emergency department in a comatose state. She has a 20-year history of alcohol abuse. She had an episode of hematemesis a day ago. Laboratory studies show elevated blood ammonia levels, low serum albumin levels and AST:ALT ratio of 2:1. Which of the following is most likely observed in this patient?

- A. Increased activity of glutaminase in brain
- B. Increased activity of glutamine synthetase in brain
- C. Increased activity of urea cycle in liver
- D. Increased formation of α -ketoglutarate from glutamate in brain
- E. Increased activity of the TCA cycle in brain

5. A 5-day-old girl is brought to the physician by her parents for a follow-up examination. Newborn screening studies show elevated levels of phenylalanine in circulation. Which of the following complications is most likely in this patient if treatment is not immediately begun?

- A. Increased accumulation of tyrosine in circulation
- B. Increased pigmentation of cartilage
- C. Lowering of IQ and intellectual disability
- D. Increased risk of forming renal stones
- E. Increased risk of metabolic acidosis

6. A 10-year-old boy is brought to the physician by his parents because of a 4-week history of tingling and numbness of his hands and feet. Laboratory studies confirm pyridoxine deficiency. Which of the following enzyme activities is most likely impaired in this patient?

- A. Glutamate dehydrogenase
- B. Tyrosine hydroxylase
- C. Methylmalonyl CoA mutase
- D. Dopa decarboxylase
- E. Homocysteine methyl transferase

7. A 46-year-old man comes to the physician because of a 1-month history of tingling and numbness in his hands and feet. He also says he has mood swings. Laboratory studies show low hemoglobin levels and MCV is $105\mu\text{m}^3$ (Normal: $80\text{-}100\mu\text{m}^3$). Which of the following additional laboratory findings is most likely in this patient?

- A. Increased serum homocysteine levels
- B. Low serum ferritin levels
- C. Increased serum methylmalonic acid levels
- D. Presence of phenylacetate and phenyllactate in urine
- E. Increased serum homogentisic acid levels

8. A researcher is studying the metabolism of serotonin in patients with Carcinoid syndrome. Which of the following amino acids is most likely a precursor of this hormone?

- A. Tyrosine
- B. Glutamate
- C. Tryptophan
- D. Histidine
- E. Glycine

9. A 52-year-old woman comes to the physician for a follow-up examination. She has been surgically treated for breast cancer a week ago. She is prescribed an anti-cancer drug that is a folate analog and inhibits the formation of active tetrahydrofolate. Which of the following drugs is most likely prescribed to this patient?

- A. 5-fluorouracil
- B. Allopurinol
- C. Methotrexate
- D. Trimethoprim
- E. Mycophenolic acid

10. A 32-year-old man comes to the physician because of a 1-week history of cough and fever. Physical examination shows diffuse crepitations over both the lungs and a diagnosis of bronchitis is made. Laboratory studies show increased excretion of ammonium chloride in his urine. Which of the following is most likely the immediate precursor of ammonia (NH_3) in the renal tubules?

- A. Arginine
- B. Alanine

- C. Asparagine
- D. Glutamine
- E. Lysine

11. A researcher is studying amino acid pool and metabolism in various groups of patients. Which of the following best describes the metabolism of these compounds?

- A. All amino acids are glucogenic
- B. All amino acids are essential amino acids
- C. The major excretory end product of amino acid catabolism is ammonia
- D. During starvation, patients are in positive nitrogen balance
- E. Tissue protein synthesis is increased during periods of growth

12. A 25-year-old woman is admitted to the hospital due to difficulty breathing. She has a 10-day history of fever and productive cough. She is prescribed antibiotics for the lung infection. Her fever subsides and her appetite improves. Which of the following best describes nitrogen metabolism during the stage of recovery?

- A. Increased tissue protein catabolism
- B. Nitrogen intake would be equal to urinary nitrogen output
- C. Dietary protein content should be increased
- D. No net gain or loss of muscle mass
- E. State of negative nitrogen balance

13. A research group is studying the plasma glucose levels in a group of 20-year-old men on voluntary fasting. The plasma glucose level is observed after infusion of various amino acids. It is observed that infusion of a specific amino acid results in no increase in plasma glucose levels. Which of the following amino acids most likely results in this effect?

- A. Aspartate
- B. Glutamate
- C. Glutamine
- D. Leucine
- E. Alanine

14. A 3-month-old girl is brought to the physician for a follow-up examination. She has a history of developmental delay and intermittent seizures since birth. Laboratory studies confirm an inborn error of metabolism. Oral thiamine supplementation results in improvement of the symptoms. Which of the following is the most likely diagnosis?

- A. Classic phenylketonuria
- B. Homocystinuria
- C. Maple syrup urine disease
- D. Cystinuria
- E. Alkaptonuria
- F. Methylmalonic aciduria

15. Amino acids and their specialized products synthesized are shown. Choose the incorrect pair.

- A. Tryptophan – Serotonin
- B. Tyrosine – Glutathione
- C. Tyrosine – Epinephrine

- D. Arginine – Nitric oxide
- E. Glycine – Heme
- F. Glutamic acid – GABA

16. Inborn errors of amino acid metabolism and the related amino acids are paired. Choose the appropriate pair.

- A. Maple syrup urine disease – Tryptophan
- B. Phenylketonuria – Branched chain amino acids
- C. Methylmalonic aciduria – Valine
- D. Albinism – Arginine
- E. Hartnup Disease – Cystine
- F. Alkaptonuria – Methionine

17. Inherited metabolic disorders are paired with the enzymes deficient. Choose the inappropriate pair.

- A. Hyperammonemia I – Carbamoyl Phosphate Synthase I
- B. Phenylketonuria – Phenylalanine hydroxylase
- C. Albinism – Tyrosinase
- D. Maple syrup urine disease – Methylmalonyl CoA mutase
- E. Homocystinuria – Cystathionine synthase

18. A 60-year-old woman comes to the physician because of a 2-month history of resting tremors. Physical examination shows muscle rigidity and resting tremors. A diagnosis of Parkinson's disease is made and DOPA is prescribed. Which of the following enzymes is most likely required for the formation of the active neurotransmitter from this drug?

- A. Dopamine β hydroxylase
- B. Dopa decarboxylase
- C. Phenylethanolamine N-methyl transferase
- D. Tyrosine hydroxylase
- E. Monoamine oxidase

19. A 5-day-old girl is brought to the physician by her parents for a follow-up examination. She has had a newborn screening test which showed hyperphenylalaninemia. Further laboratory tests show a deficiency of dihydrobiopterin synthesis. Which of the following reactions is most likely impaired in the patient?

(More than one correct answer)

- A. Formation of GABA from glutamate
- B. Formation of serotonin from tryptophan
- C. Formation of histamine from histidine
- D. Conversion of tyrosine to dihydroxyphenylalanine
- E. Conversion of creatine to phosphocreatine
- F. Formation of epinephrine from norepinephrine
- G. Conversion of phenylalanine to tyrosine

20. A 25-year-old woman comes to the physician because of a 2-month history of feeling tired. She says she is on a diet which does not include fruits and vegetables. Physical examination shows conjunctival pallor. Laboratory studies show macrocytic anemia and low serum folate levels. Which of the following is most likely impaired in the patient?

- A. Conversion of homocysteine to cystathionine
- B. Conversion of methylmalonyl CoA to succinyl CoA

- C. Synthesis of thymidine monophosphate
- D. Conversion of phenylalanine to tyrosine
- E. Synthesis of serotonin

21. A 57-year-old man comes to the physician because of a 6-month history of tingling and numbness in his hands and feet. Physical examination shows conjunctival pallor. Laboratory studies show macrocytic anemia and low serum vitamin B12 levels. Which of the following additional laboratory findings is most likely in this patient?

- A. Increased propionate levels
- B. Decreased serum iron levels
- C. Increased homocysteine levels
- D. Decreased levels of serum folate
- E. Increased levels of methylmalonate

22. Match the following inherited disorders (A-F) with the observed clinical or laboratory finding (I - VIII):

- A. Albinism
- B. Argininosuccinic aciduria
- C. Phenylketonuria
- D. Homocystinuria
- E. Maple syrup urine disease
- F. Alkaptonuria

- I. Symptoms can be improved with vitamin B12 supplementation
- II. Decreased formation of Urea
- III. Decreased conversion of homocysteine to cysteine
- IV. Decreased catabolism of tyrosine
- V. Decreased activity of tyrosinase
- VI. Increased tryptophan excretion in urine
- VII. Decreased synthesis of Tyrosine
- VIII. Decreased catabolism of the branched chain amino acids

23. A 12-day-old boy is brought to the emergency department by his parents because of a 1-day history of poor feeding and lethargy. Laboratory studies show elevated serum ammonia levels and high levels of circulating citrulline. Which of the following inherited enzyme deficiency disorders is most likely in this patient?

- A. Carbamoyl phosphate synthetase-I
- B. Ornithine transcarbamoylase
- C. Argininosuccinate synthetase
- D. Argininosuccinate lyase
- E. Arginase

24. A 48-year-old man is brought to the emergency department in an unconscious state. He has a 2-year history of liver cirrhosis and a diagnosis of acute fulminant hepatocellular failure is made. Which of the following laboratory parameters would most likely be elevated in this patient?

- A. Blood urea nitrogen levels
- B. Blood creatinine levels

- C. Blood ammonia levels
- D. Blood albumin levels
- E. Blood uric acid levels

25. A 3-month-old girl is brought to the physician for a follow-up examination. She has a history of developmental delay. Laboratory studies show a loss of function mutation in the enzyme dihydrobiopterin reductase. Which of the following compounds will most likely be produced at reduced levels in this patient?

- A. Bilirubin
- B. Palmitic acid
- C. Epinephrine
- D. Heme
- E. Histamine

26. A 26-year-old woman is brought to the emergency department in a drowsy state. She has a 10-year history of type 1 diabetes mellitus and is on insulin therapy. Laboratory studies show metabolic acidosis and elevated urinary ammonium excretion. Which of the following compounds most likely undergoes enzymatic hydrolysis to form this urinary metabolite?

- A. Glutamine
- B. Glutamic acid
- C. Histamine
- D. Tyrosine
- E. Urea

27. A 5-year-old boy comes to the physician for a follow-up examination. He had been diagnosed with deficiency in aldolase B. Laboratory studies show hyperuricemia. Which of the following disorders is most likely in this patient?

- A. Hereditary fructose intolerance
- B. von Gierke disease
- C. Hyperammonemia type-II
- D. Benign fructosuria
- E. McArdle disease
- F. Classical galactosemia

28. An investigator is studying red blood cells which rely exclusively on glycolysis for the generation of ATP. ATP is necessary for the maintenance of the plasma membrane voltage potential and osmotic integrity. It is observed that these cells maintain a higher level of K^+ than of Na^+ inside the cell. Which of the following best describes this observation?

- A. The Na^+ leaks out into the surrounding extracellular fluid.
- B. K^+ is bound to sites inside the cell.
- C. The body contains less total Na^+ than K^+ , so Na^+ is found at a low concentration inside the cell.
- D. Na^+ is actively transported out of the cell and K^+ actively transported into the cell
- E. Much of the Na^+ outside the cell is bound to protein, so that no Na^+ gradient exists across the cell membrane.

29. A 2-day-old boy is brought to the emergency department because of progressively worsening lethargy and refusal to feed. His respiration is increased. Physical examination shows response only to painful stimuli. Laboratory studies show mild respiratory alkalosis. Blood samples are sent to the laboratory for metabolite analysis. Laboratory studies show an elevated level of ammonia in the blood. Which of the following amino acids is also most likely to be elevated in his blood?

- A. Glutamine
- B. Arginine
- C. Lysine
- D. Leucine
- E. Glycine

30. A 25-year-old man is brought to the emergency department because of an automobile accident. He was stranded in the mountains for 9 days. He had found shelter and water, but he did not have access to food. During this time, which of the following amino acids were catabolized for the exclusive use of ketone body synthesis?

- A. Lysine
- B. Isoleucine
- C. Tyrosine
- D. Glycine
- E. Alanine

31. A 45-year-old man comes to the physician with a 1-day history of decreased urination. He has a 3-day history of swelling of his feet. His blood pressure is 150/90mmHg. Physical examination shows bilateral pitting edema. Laboratory studies confirm acute renal disease. Which of the following parameters would be the best indicators of renal function in this patient?

- A. Serum ammonia
- B. Serum albumin
- C. Blood urea nitrogen
- D. Blood glucose
- E. Serum free fatty acids

32. A 12-year-old girl is brought to the physician by her parents because of a 2-month history of diarrhea and skin rashes on her arms. Her 8-year-old sister has a similar history but milder clinical features. Physical examination shows a red scaly rash on her arms and neck. Neurological examination shows mild cerebellar ataxia. Dietary history shows adequate protein and niacin. Urinalysis shows elevated amounts of certain free amino acids. Which of the following amino acids is most likely found in her urine?

- A. Glutamine
- B. Lysine
- C. Cystine
- D. Tryptophan
- E. Glycine

33. A researcher isolates an enzyme obtained from liver biopsy specimens. It is observed that the enzyme forms free ammonia. Which of the following enzymes is most likely isolated by this researcher?

- A. Alanine aminotransferase
- B. Aspartate aminotransferase

- C. Glutamate dehydrogenase
- D. Glutamine synthetase
- E. Asparagine synthetase

34. A researcher is studying ammonia metabolism in the brain in experimental animals with liver disease. It is observed that ammonia is used and detoxified for the formation of an amino acid in the nervous tissue of these animals. Which of the following enzymes is most significant for performing this function in the brain?

- A. Alanine aminotransferase
- B. Aspartate aminotransferase
- C. Glutamate dehydrogenase
- D. Glutamine synthetase
- E. Glutaminase

35. A 32-year-old man comes to the physician because of a 2-month history of memory loss. He has been diagnosed with depression and has been drinking excessively for over one year. His major source of calories is ethanol, and he rarely eats well-balanced healthy meals. Which of the following enzyme activities is most likely impaired in this patient?

- A. Propionyl CoA carboxylase
- B. Glutamate dehydrogenase
- C. Pyruvate dehydrogenase
- D. Methionine synthase
- E. DOPA decarboxylase

List some coenzymes important in amino acid metabolism and list at least one reaction requiring that coenzyme

Coenzyme	Enzyme requiring
PLP (Vitamin B6)	
Tetrahydrobiopterin (BH4)	
TPP (Vitamin B1)	
NAD ⁺ (Niacin; Vitamin B3)	
Vitamin B ₁₂ (Cobalamin)	

List specialized products of the amino acids:

Amino acid	Specialized product
Phenylalanine, Tyrosine	
Tryptophan	
Arginine	
Glutamic acid	
Histidine	
Glycine	

Answer Key:

1. D
2. B
3. C
4. B
5. C
6. D
7. C
8. C
9. C
10. D
11. E
12. C
13. D
14. C
15. B
16. C
17. D
18. B
19. B, D, G
20. C
21. E
22. **Answer:** A – V; B – II; C – VII; D – III; E – VIII; F – IV
23. C
24. C
25. C
26. A
27. A

- 28. D
- 29. A
- 30. A
- 31. C
- 32. D
- 33. C
- 34. D
- 35. C

Short Explanations:

1. A 10-year-old boy is brought to the physician by his parents because of a 3-week history of burning sensation during urination. Urinalysis shows presence of red blood cells and hexagonal crystals. Which of the following is an additional finding in this patient?

Comment: This is a patient with cystinuria. Note the hematuria (RBC in urine) and hexagonal cystine crystals.

- A. Decreased tubular reabsorption of tryptophan – Occurs in Hartnup disease; Defect in the tryptophan transporter in GIT and renal tubule; results in less absorption of dietary tryptophan and increased tryptophan losses in urine. Patients have poor growth (due to deficiency of tryptophan, an essential amino acid), features of pellagra (Niacin deficiency; NAD synthesis requires tryptophan).
- B. Deficiency of cystathionine β -synthase – Results in inherited homocystinuria; A defect in conversion of homocysteine to cysteine; Patients present with skeletal deformities, lens dislocation, and accelerated atherosclerosis. PLP is the coenzyme
- C. Increased circulating levels of homocysteine – Observed in inherited homocystinuria; May also be observed in folate, vitamin B12 and pyridoxine deficiency. Associated with increased risk of cardiovascular disease
- D. **Increased urinary excretion of cystine – Correct answer** – Characterized by defect in the amino acid transporter for Cystine, arginine, ornithine and lysine (COAL) in renal tubules. There is decreased reabsorption of cystine and increased cystine loss in urine. Cystine has a low solubility and precipitates to form renal stones.
- E. Elevated levels of homogentisic acid in urine

2. A 1-month-old boy is brought to the physician by his parents for a follow-up examination. Newborn screening studies showed elevated phenylalanine levels in blood and urine. He was prescribed a phenylalanine restricted diet. Physical examination shows a delay in the developmental milestones. He is prescribed sapropterin (biopterin analog). Which of the following reactions most likely requires this coenzyme?

Comment: This is most likely a patient with PKU-2 (PKU-II) - or Malignant PKU. It may be caused by a deficiency in BH4 ... there are two ways to be deficient in BH4:

- 1. Patients can't make the BH2 (deficiency in BH2 synthesis). Note that synthesis of the pterin ring (BH2) may be accomplished in mammals, so BH2 is NOT a vitamin. BH2 is not a required component in the diet of a normal person

2. Patients can't recycle BH₂. They can't reduce BH₂ back to BH₄ after the PAH enzyme converts PHE to TYR. This would be due to a deficiency in dihydropteridine reductase
 PKU 2 is worse (malignant) because there are two negative effects - PHE accumulation, AND decreased synthesis of catecholamines AND serotonin which are essential neurotransmitters in the brain and hormones in the body

A. Conversion of DOPA to dopamine – The enzyme DOPA decarboxylase uses PLP (vitamin B₆) as a coenzyme

B. **Conversion of Tyrosine to DOPA – Correct answer** – Tetrahydrobiopterin is required for three reactions: Phenylalanine → Tyrosine; Tyrosine → DOPA (DOPA later forms dopamine, norepinephrine and epinephrine) and Tryptophan → 5-hydroxytryptophan (Later forms serotonin)

C. Conversion of 5-hydroxy tryptophan to serotonin – This reaction is catalyzed by amino acid decarboxylase which requires PLP (vitamin B₆)

D. Melanin synthesis – Melanin is synthesized from tyrosine; Melanin synthesis may be impaired in PKU due to inhibition of tyrosinase (enzyme for melanin synthesis) by elevated levels of phenylalanine

E. Conversion of branched chain amino acids to branched chain keto acids – Catalyzed branched chain alpha keto acid dehydrogenase. It requires NAD, FAD, TPP, CoA and lipoic acid as coenzymes. Deficient in maple syrup urine disease (MSUD). Thiamine (Vitamin B₁) supplements are useful in MSUD.

3. A research group is studying the effects of a diet in a population that subsists on a protein source that is deficient in methionine. Synthesis of which of the following amino acids is most likely impaired in these subjects?

Comment: Methionine (sulfur containing essential amino acid) is a precursor for cysteine (sulfur containing non-essential amino acid) synthesis. Methionine → Homocysteine → Cystathionine → Cysteine. This pathway requires vitamin B₆ as PLP. Cysteine synthesis depends on dietary methionine availability. Cysteine is a conditionally essential amino acid.

Essential amino acids are: Methionine, Tryptophan, Histidine, Isoleucine, Leucine, Lysine, Phenylalanine, Valine, Threonine (MTHILL PVT or PVT TIMHLL); Lippincott's Biochemistry: Fig 20.2

	Glucogenic	Glucogenic and Ketogenic	Ketogenic
Nonessential	Alanine Arginine Asparagine Aspartate Cysteine Glutamate Glutamine Glycine Proline Serine	Tyrosine	
Essential	Histidine Methionine Threonine Valine	Isoleucine Phenylalanine Tryptophan	Leucine Lysine

A. Alanine – is a non-essential amino acid. Formed from pyruvate

B. Tyrosine – is a conditionally essential amino acid. Formed from phenylalanine (essential amino acid)

C. **Cysteine – Correct answer** – Formed from methionine

D. Phenylalanine – is an essential amino acid and cannot be synthesized in humans. It has to be provided in the diet

E. Glycine – is a non-essential amino acid and can be synthesized in humans

4. A 55-year-old woman is brought to the emergency department in a comatose state. She has a 20-year history of alcohol abuse. She had an episode of hematemesis a day ago. Laboratory studies show elevated blood ammonia levels, low serum albumin levels and AST:ALT ratio of 2:1. Which of the following is most likely observed in this patient?

Comment: The laboratory findings and presentation are suggestive of chronic alcoholic cirrhosis with hepatic encephalopathy. The elevated blood ammonia levels result in increased conversion of alpha-keto glutarate to glutamate and increased glutamine synthesis.

- A. Increased activity of glutaminase in brain – There would be increased synthesis of glutamine from glutamate. Note glutaminase is involved in the formation of glutamate from glutamine with the liberation of ammonia
- B. **Increased activity of glutamine synthetase in brain – Correct answer – Increased ammonia levels increases the synthesis of glutamine in the brain by increasing activity of glutamine synthetase**
- C. Increased activity of urea cycle in liver – The blood ammonia levels are increased because of impaired urea cycle due to the chronic liver cirrhosis.
- D. Increased formation of α -ketoglutarate from glutamate in brain – High blood ammonia levels results in increased glutamate formation from alpha ketoglutarate via glutamate dehydrogenase.
- E. Increased activity of the TCA cycle in brain – Low levels of alpha ketoglutarate can result in low activity of the Krebs cycle. Results in impaired energy generation in the brain.

5. A 5-day-old girl is brought to the physician by her parents for a follow-up examination. Newborn screening studies show elevated levels of phenylalanine in circulation. Which of the following complications is most likely in this patient if treatment is not immediately begun?

Comment: This is a child with PKU-I (more common) which is characterized by deficiency of phenylalanine hydroxylase. PKU-II (less common) also has similar findings. PKU-II is characterized by deficiency of the coenzyme tetrahydrobiopterin. Laboratory studies show increased phenyllactate and phenylpyruvate in blood and urine.

- A. Increased accumulation of tyrosine in circulation – Low activity of phenylalanine hydroxylase results in low tyrosine levels. Tyrosine levels are elevated in children with tyrosinemia
- B. Increased pigmentation of cartilage – and joint problems are complications of alkaptonuria due to accumulation of homogentisic acid. Characterized by deficiency of homogentisate oxidase (enzyme involved in tyrosine degradation)
- C. **Lowering of IQ and intellectual disability – Correct answer - Children with untreated PKU have developmental delay and intellectual disability. Treatment is dietary restriction of phenylalanine and providing tyrosine in the diet. Early treatment results in better prognosis and improvement in neurological function. Lifelong treatment is usually required.**
- D. Increased risk of forming renal stones – Cystinuria is characterized by occurrence of renal stones. In patients with cystinuria there is deficiency of the renal transporter for cystine, ornithine, arginine and lysine. This results in increased cystine excretion in urine and formation of renal stones due to high urinary cystine levels.

E. Increased risk of metabolic acidosis – Metabolic acidosis is observed in the amino acid disorders that produce weak acids like methylmalonic aciduria (methylmalonate); maple syrup urine disease (presence of branched chain keto acids). Pyruvate dehydrogenase deficiency (Accumulation of lactate), von Gierke disease can also result in metabolic acidosis.

6. A 10-year-old boy is brought to the physician by his parents because of a 4-week history of tingling and numbness of his hands and feet. Laboratory studies confirm pyridoxine deficiency. Which of the following enzyme activities is most likely impaired in this patient?

Comment: Pyridoxine (vitamin B6) forms PLP as a coenzyme. PLP is required for aminotransferases, amino acid decarboxylases, homocysteine metabolism (cystathionine synthase) and heme synthesis (delta-ALA synthase)

A. Glutamate dehydrogenase – Uses NAD as a coenzyme. It converts alpha-ketoglutarate to glutamate using ammonia (ammonia fixation). It can also form ammonia in the reverse direction when glutamate is converted to alpha-ketoglutarate. The ammonia formed can next be detoxified in the urea cycle in the liver.

B. Tyrosine hydroxylase – Uses tetrahydrobiopterin as a coenzyme. This enzyme converts tyrosine to dihydroxyphenylalanine (DOPA) for the synthesis of the catecholamines

C. Methylmalonyl CoA mutase – Uses vitamin B12 as a coenzyme. This enzyme converts methylmalonyl CoA to succinyl CoA which can be used via the Krebs cycle. Methylmalonyl CoA is formed from Propionyl CoA by propionyl CoA carboxylase which requires biotin as a coenzyme. Propionyl CoA is formed by metabolism of the branched chain amino acids and odd chain fatty acids. This enzyme is deficient in methylmalonic aciduria.

D. **Dopa decarboxylase – Correct answer** – Uses PLP as a coenzyme. Converts DOPA to dopamine which later forms norepinephrine and epinephrine (catecholamines). 5-hydroxytryptophan decarboxylase (serotonin synthesis) also uses PLP. PLP is also required for the synthesis many other neurotransmitters by amino acid decarboxylases.

E. Homocysteine methyl transferase – Also known as methionine synthase. Uses vitamin B12 and methyl-tetrahydrofolate (vitamin B9) as coenzymes. This enzyme converts homocysteine to methionine (recycling of methionine). Conversion of homocysteine to cysteine (cystathionine synthase) requires PLP (vitamin B6). Hence optimum metabolism of homocysteine requires vitamin B12, folic acid and pyridoxine.

7. A 46-year-old man comes to the physician because of a 1-month history of tingling and numbness in his hands and feet. He also says he has mood swings. Laboratory studies show low hemoglobin levels and MCV is $105\mu\text{m}^3$ (Normal: $80\text{--}100\mu\text{m}^3$). Which of the following additional laboratory findings is most likely in this patient?

Comment: The vignette describes a patient with neurological and psychiatric manifestations and macrocytic anemia (Increased MCV). This indicates a patient with vitamin B12 deficiency. An important differential diagnosis is folate deficiency which results in macrocytic anemia but no neurological manifestations.

A. Increased serum homocysteine levels – Is observed in both vitamin B12 and folate deficiency. Not the best answer! Methionine synthase uses both vitamin B12 and methyltetrahydrofolate; hence homocysteine levels

are elevated in both B12 and folate deficiency. Homocysteine metabolism (cystathionine synthase) requires PLP and homocysteine levels are elevated in B6 deficiency too; but patients with B6 deficiency have microcytic (low MCV) anemia.

B. Low serum ferritin levels – observed in iron deficiency anemia. Iron deficiency anemia is characterized by microcytic anemia (low MCV), low serum ferritin, low serum iron levels, decreased percent saturation and increased TIBC.

C. **Increased serum methylmalonic acid levels – Correct answer** – Elevated methylmalonate levels are specifically found in vitamin B12 deficiency. Accumulation of methylmalonate is responsible for the neurological manifestations due to defective myelination in vitamin B12 deficiency.

D. Presence of phenylacetate and phenyllactate in urine – Phenyllactate and phenylacetate are minor products formed from phenylalanine metabolism. In patients with PKU, phenylalanine enters minor metabolic pathways and forms these intermediates which are found in high levels in blood and urine.

E. Increased serum homogentisic acid levels – Is found in patients with alkaptonuria (homogentisate oxidase deficiency). Homogentisic acid accumulates in joints and results in arthritis and black discoloration of cartilage of ears and nose (ochronosis)

8. A researcher is studying the metabolism of serotonin in patients with Carcinoid syndrome. Which of the following amino acids is most likely a precursor of this hormone?

Comment: Carcinoid tumor is a tumor of the APUD cells in the GIT. There is excessive formation of serotonin in these patients. Serotonin is metabolized to 5-hydroxyindoleacetic acid (5-HIAA). Serotonin is formed from tryptophan.

A. Tyrosine – is a precursor of catecholamines, melanin

B. Glutamate – forms glutamine; glutamate is also a precursor of GABA (neurotransmitter)

C. **Tryptophan – Correct answer** – is a precursor of serotonin (5-HT) and melatonin (pineal gland)

D. Histidine – is a precursor of histamine

E. Glycine – is used for heme synthesis

9. A 52-year-old woman comes to the physician for a follow-up examination. She has been surgically treated for breast cancer a week ago. She is prescribed an anti-cancer drug that is a folate analog and inhibits the formation of active tetrahydrofolate. Which of the following drugs is most likely prescribed to this patient?

A. 5-fluorouracil – is a pyrimidine analog; Inhibits thymidylate synthase (Pyrimidine nucleotide synthesis)

B. Allopurinol – inhibits xanthine oxidase; used in patients with hyperuricemia and gout

C. **Methotrexate – Correct answer** - is a folate analog; Competitively inhibits eukaryotic dihydrofolate reductase and is used as an anticancer agent. It inhibits purine and pyrimidine nucleotide synthesis and hence DNA synthesis and cell division.

D. Trimethoprim – Inhibits bacterial dihydrofolate reductase and is used as an antibacterial agent. It inhibits purine and pyrimidine nucleotide synthesis and hence bacterial DNA synthesis

E. Sulfonamide – Inhibits bacterial folic acid synthesis and is used as an antibacterial agent. It inhibits purine and pyrimidine nucleotide synthesis and hence bacterial DNA synthesis. Remember, humans do NOT synthesize folic acid and it is obtained in the diet.

10. A 32-year-old man comes to the physician because of a 1-week history of cough and fever. Physical examination shows diffuse crepitations over both the lungs and a diagnosis of bronchitis is made. Laboratory studies show increased excretion of ammonium chloride in his urine. Which of the following is most likely the immediate precursor of ammonia (NH_3) in the renal tubules?

Comment: The patient is most likely in a state of chronic respiratory acidosis. Ammonium in the urine is increased due to increased secretion of protons in the renal tubule (renal compensation for respiratory acidosis). Ammonia is formed in the renal tubule by the action of glutaminase.

- A. Arginine
- B. Alanine
- C. Asparagine
- D. **Glutamine – Correct answer** – About 60% of the ammonia in the renal tubule is formed by hydrolysis of glutamine by glutaminase
- E. Lysine

11. A researcher is studying amino acid pool and metabolism in various groups of patients. Which of the following best describes the metabolism of these compounds?

- A. All amino acids are glucogenic – Some amino acids are both glucogenic and ketogenic and two amino acids (Leucine and lysine) are only ketogenic
- B. All amino acids are essential amino acids – 8 of the 20 amino acids are essential; 2 are semi-essential during periods of growth and positive nitrogen balance
- A. The major excretory end product of amino acid catabolism is ammonia – Urea is the major excretory product of amino acid catabolism
- C. During starvation, patients are in positive nitrogen balance – During starvation, patients are in negative nitrogen balance. The intake of nitrogen (protein) in the diet is less than the loss of nitrogen (urea) in urine and there is increased breakdown of tissue protein. Protein breakdown forms amino acids that are used for gluconeogenesis to maintain plasma glucose levels
- D. **Tissue protein synthesis is increased during periods of growth - Correct answer** – Infancy, childhood, adolescence, pregnancy and recovery from illness are examples of positive nitrogen balance

12. A 25-year-old woman is admitted to the hospital due to difficulty breathing. She has a 10-day history of fever and productive cough. She is prescribed antibiotics for the lung infection. Her fever subsides and her appetite improves. Which of the following best describes nitrogen metabolism during the stage of recovery? Comment: During recovery (anabolic phase), tissue protein synthesis is increased and patients are in positive nitrogen balance.

- A. Increased tissue protein catabolism – Tissue protein synthesis is increased (anabolic phase)
- B. Nitrogen intake would be equal to urinary nitrogen output – This is a state of nitrogen equilibrium which is observed in healthy adults

- C. **Dietary protein content should be increased – Correct answer** – To promote anabolism and recovery of muscle mass after an acute infection, dietary protein content is increased. During recovery, there is positive nitrogen balance. Increased dietary protein and activity promote anabolism during the recovery phase
- D. No net gain or loss of muscle mass – There is increased tissue protein synthesis during recovery (anabolism).
- E. State of negative nitrogen balance – During the illness there is negative nitrogen balance, but during recovery the patient is in positive nitrogen balance

13. A research group is studying the plasma glucose levels in a group of 20-year-old men on voluntary fasting. The plasma glucose level is observed after infusion of various amino acids. It is observed that infusion of a specific amino acid results in no increase in plasma glucose levels. Which of the following amino acids most likely results in this effect?

Comment: The key here is to identify the amino acid which cannot form glucose (an amino acid not used in gluconeogenesis). Leucine and lysine are only ketogenic amino acids. Note that the other amino acids are glucogenic.

- A. Aspartate – forms oxaloacetate via aspartate aminotransferase and is used for gluconeogenesis
- B. Glutamate – forms alpha ketoglutarate via glutamate dehydrogenase or aminotransferase and is used for gluconeogenesis
- C. Glutamine - forms glutamate via glutaminase which later forms alpha ketoglutarate and is used for gluconeogenesis
- D. **Leucine – Correct answer – is exclusively ketogenic amino acid**
- E. Alanine – forms pyruvate via alanine aminotransferase which is later used for glucose formation

14. A 3-month-old girl is brought to the physician for a follow-up examination. She has a history of developmental delay and intermittent seizures since birth. Laboratory studies confirm an inborn error of metabolism. Oral thiamine supplementation results in improvement of the symptoms. Which of the following is the most likely diagnosis?

- A. Phenylketonuria – Characterized by deficiency of Phenylalanine hydroxylase. Requires tetrahydrobiopterin as a coenzyme. Sapropterin (synthetic analog of tetrahydrobiopterin) has been used in some patients with PKU-I and in all patients with PKU-II. 'Musty' or 'mousy' odor of urine.
- B. Homocystinuria – Characterized by deficiency of cystathionine synthase. Requires vitamin B6 (PLP)
- C. **Maple syrup urine disease – Correct answer** – Characterized by deficiency of branched alpha-keto acid dehydrogenase. Urine 'burnt sugar' or 'maple syrup' odor. Management by restriction of branched chain amino acids (Leucine, isoleucine and valine) in diet. Requires vitamin B1 (TPP) as a coenzyme. Laboratory tests show branched chain keto acids in urine (responsible for the urine odor)
- D. Cystinuria – Characterized by deficiency of the amino acid transporter for cystine, ornithine, arginine and lysine. Presents with renal stones containing cystine.
- E. Alkaptonuria – Deficiency of homogentisate oxidase; Presents in older patients (30-50 years) as ochronosis and arthritis due to accumulation of homogentisic acid in cartilage.
- F. Methylmalonic aciduria – Deficiency of methylmalonyl CoA mutase; requires vitamin B12 as a coenzyme

15. Amino acids and their specialized products synthesized are shown. Choose the **incorrect pair**.

Comment: This question is a quick refresher regarding amino acids and the compounds formed from them.

- A. Tryptophan – Serotonin; Precursor of serotonin and melatonin
- B. **Tyrosine – Glutathione; Correct answer - Glutathione an antioxidant contains glycine, glutamate and cysteine**
- C. Tyrosine – Epinephrine; Tyrosine is the precursor of catecholamines which are dopamine, epinephrine and norepinephrine. It is also a precursor of melanin.
- D. Arginine – Nitric oxide; by nitric oxide synthase
- E. Glycine – Heme; Glycine and succinyl CoA form delta-amino levulinic acid (Delta-ALA) which eventually forms heme
- F. Glutamic acid – GABA; by glutamate decarboxylase

16. Inborn errors of amino acid metabolism and the related amino acids are paired. Choose the **appropriate** pair.

- A. Maple syrup urine disease – Tryptophan; Defect in metabolism of branched chain amino acids, valine, leucine and isoleucine
- B. Phenylketonuria – Branched chain amino acids; Defect in phenylalanine catabolism to tyrosine
- C. **Methylmalonic aciduria – Valine ; Correct answer: Valine and isoleucine (branched chain amino acids) and odd chain fatty acids form propionyl CoA which is first converted to methylmalonyl CoA and then to succinyl CoA.**
- D. Albinism – Arginine; Albinism is due to deficiency of tyrosinase required for synthesis of melanin
- E. Hartnup Disease – Cystine; Hartnup disease is characterized by deficiency of the amino acid transporter for tryptophan. Results in reduced dietary tryptophan absorption and excretion of tryptophan in urine
- F. Alkaptonuria – Methionine; Alkaptonuria is due to deficiency of homogentisate oxidase required for catabolism of tyrosine.

17. Inherited metabolic disorders are paired with the enzymes deficient. Choose the **inappropriate** pair.

- A. Hyperammonemia I – Carbamoyl Phosphate Synthase I
- B. Phenylketonuria – Phenylalanine hydroxylase
- C. Albinism – Tyrosinase; results in light colored hair, eyes and skin; photosensitivity and photophobia
- D. **Maple syrup urine disease – Methylmalonyl CoA mutase; Maple syrup urine disease is caused by deficiency of branched chain alpha keto acid dehydrogenase**
- E. Homocystinuria – Cystathionine synthase

18. A 60-year-old woman comes to the physician because of a 2-month history of resting tremors. Physical examination shows muscle rigidity and resting tremors. A diagnosis of Parkinson's disease is made and DOPA is prescribed. Which of the following enzymes is most likely required for the formation of the active neurotransmitter from this drug?

- A. Dopamine β hydroxylase – Converts dopamine to norepinephrine; Requires vitamin C
- B. **DOPA decarboxylase – Correct answer - Converts DOPA to dopamine. Requires PLP as a coenzyme**
- C. Phenylethanolamine N-methyl transferase – Converts norepinephrine to epinephrine; Requires SAM
- D. Tyrosine hydroxylase – forms DOPA from tyrosine; Requires tetrahydrobiopterin as a coenzyme
- E. Monoamine oxidase – Inactivates Epinephrine and norepinephrine to Vanillyl mandelic acid. Requires the enzymes MAO and COMT for the conversion.

19. A 5-day-old girl is brought to the physician by her parents for a follow-up examination. She has had a newborn screening test which showed hyperphenylalaninemia. Further laboratory tests show a deficiency of dihydrobiopterin synthesis. Which of the following reactions is most likely impaired in the patient?

(More than one correct answer)

Comment: Note that three reactions require tetrahydrobiopterin as a coenzyme; Tryptophan hydroxylase, Phenylalanine hydroxylase and tyrosine hydroxylase. The three reactions indicated in green are in a sense very much the same. All involve hydroxylation of an aromatic ring. Very similar reaction, and the same cofactor (BH₄) is used. The patient has PKU-II.

- A. Formation of GABA from glutamate – requires Glutamate decarboxylase and PLP
- B. **Formation of serotonin from tryptophan – Correct answer – Requires tryptophan hydroxylase and forms 5-hydroxytryptophan; it is next converted to 5-hydroxytryptamine (serotonin) by tryptophan decarboxylase which requires PLP as a coenzyme**
- C. Formation of histamine from histidine – Requires histidine decarboxylase and PLP
- D. **Conversion of tyrosine to dihydroxyphenylalanine (DOPA) – Correct answer – By tyrosine hydroxylase and requires tetrahydrobiopterin**
- E. Conversion of creatine to phosphocreatine – Uses ATP and creatine phosphokinase (CK)
- F. Formation of epinephrine from norepinephrine – By PNMT and requires SAM
- G. **Conversion of phenylalanine to tyrosine – Correct answer – By phenylalanine hydroxylase and requires tetrahydrobiopterin**

20. A 25-year-old woman comes to the physician because of a 2-month history of feeling tired. She says she is on a diet which does not include fruits and vegetables. Physical examination shows conjunctival pallor. Laboratory studies show macrocytic anemia and low serum folate levels. Which of the following is most likely impaired in the patient?

- A. Conversion of homocysteine to cystathionine - Requires vitamin B6 (pyridoxine)
- B. Conversion of methylmalonyl CoA to succinyl CoA – Requires vitamin B12 (cobalamin)
- C. **Synthesis of thymidine monophosphate – Correct answer - Methylene tetrahydrofolate is required for purine and thymidine synthesis and DNA synthesis during cell division. Rapidly dividing cells are mainly affected in folate deficiency**
- D. Conversion of phenylalanine to tyrosine – Requires tetrahydrobiopterin
- E. Synthesis of serotonin – Requires tetrahydrobiopterin and pyridoxine (PLP)

21. A 57-year-old man comes to the physician because of a 6-month history of tingling and numbness in his hands and feet. Physical examination shows conjunctival pallor. Laboratory studies show macrocytic anemia and low serum vitamin B12 levels. Which of the following additional laboratory findings is most likely in this patient?

Comment: Folate and B12 deficiency result in macrocytic anemia (MCV >100). B12 deficiency also has neurological (tingling and numbness) and psychiatric manifestations.

- A. Increased propionate levels – Observed in biotin deficiency; Propionyl CoA carboxylase uses biotin as a coenzyme (ABC enzyme)
- B. Decreased serum iron levels – Seen in iron deficiency anemia; Results in Microcytic anemia
- C. Increased homocysteine levels – Found in both folate and B12 deficiency and is not useful to differentiate between the two disorders
- D. Decreased levels of serum folate – Folate levels are low in folate deficiency; In B12 deficiency, there is trapping of methyltetrahydrofolate (folate trap)

E. **Increased levels of methylmalonate – Correct Answer** – Methylmalonate levels are increased in B12 deficiency and normal in folate deficiency; Methylmalonate accumulation is responsible for the neurological manifestations in patients with B12 deficiency. Vitamin B12 is required for the conversion of methylmalonyl CoA to succinyl CoA

22. Match the following inherited disorders (A-F) with the observed clinical or laboratory finding (I - VIII):

- A. Albinism
 - B. Argininosuccinic aciduria
 - C. Phenylketonuria
 - D. Homocystinuria
 - E. Maple syrup urine disease
 - F. Alkaptonuria
-
- I. Symptoms can be improved with vitamin B12 supplementation
 - II. Decreased formation of Urea
 - III. Decreased conversion of homocysteine to cysteine
 - IV. Decreased catabolism of tyrosine
 - V. Decreased activity of tyrosinase
 - VI. Increased tryptophan excretion in urine
 - VII. Decreased synthesis of Tyrosine
 - VIII. Decreased catabolism of the branched chain amino acids

22. Answer: A – V; B – II; C – VII; D – III; E – VIII; F – IV

23. A 12-day-old boy is brought to the emergency department by his parents because of a 1-day history of poor feeding and lethargy. Laboratory studies show elevated serum ammonia levels and high levels of circulating citrulline. Which of the following inherited enzyme deficiency disorders is most likely in this patient?

Comment: All the inherited urea cycle defects are characterized by high blood ammonia and glutamine levels. Name the intermediates that would accumulate in each of the listed enzyme deficiency disorders.

- A. CPS1 - links free ammonia and bicarbonate to form carbamoyl phosphate. CPS-I deficiency results in severe hyperammonemia; Hyperammonemia type 1
- B. Ornithine transcarbamoylase - links carbamoylphosphate with ornithine. Deficiency results in hyperammonemia type-2; High ammonia levels and high orotic acid (pyrimidine) levels; Mitochondrial carbamoyl phosphate is used in the cytosol for pyrimidine biosynthesis.
- C. **Argininosuccinate synthetase – Correct answer** - links aspartate with citrulline. Laboratory findings in this enzyme deficiency are high citrulline and hyperammonemia
- D. Argininosuccinate lyase - splits argininosuccinate to yield arginine and fumarate. Deficiency results in hyperammonemia, and high argininosuccinate levels
- E. Arginase - cleaves arginine to liberate urea and re-form ornithine – Deficiency characterized by hyperammonemia and elevated arginine levels

24. A 48-year-old man is brought to the emergency department in an unconscious state. He has a 2-year history of liver cirrhosis, and a diagnosis of acute fulminant hepatocellular failure is made. Which of the following laboratory parameters would most likely be elevated in this patient?

Comment: Ammonia levels would be elevated in this patient; This is acquired hyperammonemia, not an inborn error of metabolism

Note the neurological features. Ammonia is neurotoxic. Increased blood ammonia levels results in neurotransmitter imbalance (Glutamate: GABA balance) and metabolic derangements (due to depletion of alpha-ketoglutarate and impairment of Krebs cycle)

- A. Blood urea nitrogen (BUN) levels - A liver defect would lead to reduced urea formation in the liver, and elevated blood ammonia
- B. Blood creatinine levels - A kidney disorder (renal failure) may result in elevated serum creatinine, and an elevated BUN (Blood urea nitrogen)
- C. **Blood ammonia levels – Correct answer – Ammonia is detoxified to urea in the liver. In patients with liver disease, urea formation is impaired; This results in elevated blood ammonia levels (neurotoxic)**
- D. Blood albumin levels - Albumin is a serum protein. The presence of albumin in the urine might be indicative of kidney malfunction. Serum albumin may be **low** in patients with chronic liver disease
- E. Blood uric acid levels - Uric acid is a product of purine degradation; Elevated in renal failure and gout

25. A 3-month-old girl is brought to the physician for a follow-up examination. She has a history of developmental delay. Laboratory studies show a loss of function mutation in the enzyme dihydrobiopterin reductase. Which of the following compounds will most likely be produced at reduced levels in this patient?

Comment: The patient has PKU-II which is characterized by deficiency of dihydrobiopterin synthesis or impaired recycling of tetrahydrobiopterin (deficiency of dihydrobiopterin reductase)

BH4 (active form of biopterin) is needed for the hydroxylation reaction on an aromatic ring

This reaction is similar for 3 reactions which we studied:

- Phe conversion to Tyr (Phenyl alanine hydroxylase)
- Tyr conversion to L-DOPA (Tyrosine hydroxylase)
- Tryptophan conversion to 5 HydroxyTrp (Tryptophan hydroxylase) – later decarboxylation to form serotonin

After L-DOPA is formed, the next reaction is decarboxylation to dopamine; followed by oxidation to norepinephrine and then methylation (SAM) to form epinephrine.

So, if BH4 is limiting, tyrosine formation, L-DOPA formation, and catecholamine formation is also reduced

Dopamine, norepinephrine, epinephrine, and serotonin formation are all impaired, the other compounds on this list are not affected as they do not rely on an aromatic ring hydroxylation reaction

- A. Bilirubin – Heme forms bilirubin (waste product). Does not require tetrahydrobiopterin
- B. Palmitic acid – Even chain fatty acid; Fatty acid synthesis requires biotin and NADPH
- C. **Epinephrine – Correct answer – Tyrosine, Dopamine, norepinephrine, epinephrine and serotonin synthesis requires BH4**
- D. Heme – Uses Glycine, succinyl CoA and pyridoxal phosphate (PLP)
- E. Histamine – formed from histidine by decarboxylation; requires PLP

Amino acid pool

The rates of amino acid catabolism can vary depending on the state of nutrition

In a healthy adult, the rate of protein synthesis = Rate of protein catabolism. Tissue protein levels remain a constant (12kg) (Nitrogen balance)

A healthy well fed adult is in nitrogen balance (nitrogen equilibrium):

Input of amino acids (Nitrogen/ protein intake) = Output of amino acids (Nitrogen/ Urea in urine)

Amino acid pool is in a steady state

Compare and contrast the two major protein degradation systems:

<u>Ubiquitin - Proteasome</u>	<u>Lysosomal pathway</u>
• Degrades endogenous intracellular proteins • ATP dependent • Proteins are tagged by polyubiquitin • Ubiquitin is a small globular protein • Ubiquitinated protein is hydrolyzed to small peptides by the proteasome. • Amino acids formed enter the amino acid pool • Ubiquitin is recycled and reused	• Degrades extracellular proteins like plasma proteins and proteins taken up by endocytosis • Non-energy dependent degradative pathway • Degraded by lysosomal acid hydrolases • Lysosome is membrane enclosed cellular organelle • Lysosome has a lower pH than the rest of the cell

Nitrogen balance (Equilibrium)

An adult is said to be in a state of 'Nitrogen Balance'

Input of dietary nitrogen = Output of nitrogen in Urine (sweat & feces)

The amount of tissue protein is a constant (Tissue protein synthesis = Tissue protein catabolism)

Positive nitrogen balance

Intake of dietary nitrogen is greater than the output of nitrogen

There is increased tissue protein synthesis

- The body has a net increase in anabolic pathways compared to catabolic pathways
- Body mass is increasing

Growth periods: Childhood, Adolescence, Pregnancy

Negative nitrogen balance

Intake of nitrogen is less than the Output of nitrogen

Tissue protein catabolism is increased

- Increased muscle proteolysis
- Increased breakdown of tissue protein

Increased urinary excretion of urea

Negative nitrogen balance is observed in: Starvation, Chronic illness (infections), Cancer, Uncontrolled diabetes mellitus. Also observed when the diet is deficient in protein OR a deficiency of any of the essential amino acids

Glucogenic & ketogenic amino acids

Glucogenic amino acids are those amino acids which form pyruvate or any of the TCA cycle intermediates: Glycine, Alanine, Glutamate.....

Ketogenic amino acids are those amino acids that form acetyl CoA or the ketone bodies: Leucine and lysine

Both ketogenic and glucogenic amino acids are those amino acids whose carbon skeletons form pyruvate or a TCA cycle intermediate AND acetyl CoA: Phenylalanine and tyrosine, tryptophan

26. A 26-year-old woman is brought to the emergency department in a drowsy state. She has a 10-year history of type 1 diabetes mellitus and is on insulin therapy. Laboratory studies show metabolic acidosis and elevated urinary ammonium excretion. Which of the following compounds most likely undergoes enzymatic hydrolysis to form this urinary metabolite?

Comment: Glutamine – carries ammonia from liver (and other tissues) to kidneys for pH balance of the urine. Enzyme active in the renal tubule which forms ammonia is glutaminase.

You should also be able to describe the alanine cycle between the muscle and the liver, and its importance in gluconeogenesis, and for the muscle to generate energy

- A. Glutamine – Correct answer – Glutamine forms ammonia in the renal tubules by the enzyme glutaminase; Ammonia binds to proton in the tubular lumen and forms ammonium which is later lost as ammonium chloride in urine. Acidosis stimulates glutaminase and increases urinary ammonium excretion (compensation); About 60% of urinary ammonium is formed by glutamine hydrolysis**
- B. Glutamic acid – Glutamate dehydrogenase forms ammonia from glutamic acid (mainly in liver) and provides ammonia for the urea cycle
- C. Histamine – formed from histidine; Histamine is a mediator of allergic reactions
- D. Tyrosine
- E. Urea – Major end product of nitrogen metabolism; Urinary urea is an indicator of nitrogen output

27. A 5-year-old boy comes to the physician for a follow-up examination. He had been diagnosed with deficiency in aldolase B. Laboratory studies show hyperuricemia. Which of the following disorders is most likely in this patient?

Comment: Patient has Fructose intolerance. Aldolase B deficiency leads to secondary hyperuricemia, because phosphate is sequestered into fructose as fructose-1-phosphate. This leads to many problems, but also to elevated AMP levels. AMP (purine) degradation may then lead to the presentation of hyperuricemia.

- A. Hereditary fructose intolerance – Correct answer – Patients present with hypoglycemia, liver problems, jaundice; and sometimes hyperuricemia; Presence of a reducing sugar in urine**
- B. von Gierke disease – deficiency of glucose 6-phosphatase; Patients have hypoglycemia, hepatomegaly, lactic acidosis and sometimes hyperuricemia (due to increased entry of glucose 6-phosphate into pentose pathway → Ribose 5-phosphate → Increased purine synthesis → Increased purine degradation → increased blood uric acid levels; Also lactic acid interferes with excretion of uric acid)

- C. Hyperammonemia type-II – Deficiency of Ornithine transcarbamylase; Patients have neurological problems due to hyperammonemia; Orotic acid (pyrimidine) levels increase.
- D. Benign fructosuria – deficiency of fructokinase; Asymptomatic; Present with reducing sugar in urine
- E. McArdle disease – deficiency of muscle glycogen phosphorylase; Present with cramps on intense (anaerobic) exercise; No increase in blood lactate levels following exercise
- F. Classical galactosemia – Triad of liver disease, cataracts and developmental delay; Deficiency of galactose 1-phosphate uridyl transferase (GALT); Presence of reducing sugar (galactose) in urine

28. A investigator is studying red blood cells which rely exclusively on glycolysis for the generation of ATP. ATP is necessary for the maintenance of the plasma membrane voltage potential and osmotic integrity. It is observed that these cells maintain a higher level of K^+ than of Na^+ inside the cell. Which of the following best describes this observation?

Comment: This question is related to amino acid uptake by the brush border membrane. Most of the amino acid transporters use a secondary active transport system that relies on the sodium gradient established by the Na/K pump described in this question. The sodium-glucose symporter (SGLT-1) in the intestinal mucosal cell and in the renal tubules also rely on the Na^+ gradient for uptake of glucose into these cells.

- A. The Na^+ leaks out into the surrounding extracellular fluid.
- B. K^+ is bound to sites inside the cell.
- C. The body contains less total Na^+ than K^+ , so Na^+ is found at a low concentration inside the cell.
- D. Na^+ is actively transported out of the cell and K^+ actively transported into the cell – Correct Answer –**
The Na^+-K^+ ATPase transports Na^+ out of the cell and simultaneously transports K^+ into the cell with the use of ATP (Primary active transport). The Na^+ gradient (low Na^+ levels in ICF) drives the movement of amino acids into the cells
- E. Much of the Na^+ outside the cell is bound to protein, so that no Na^+ gradient exists across the cell membrane.

29. A 2-day-old boy is brought to the emergency department because of progressively worsening lethargy and refusal to feed. His respiration is increased. Physical examination shows response only to painful stimuli. Laboratory studies show mild respiratory alkalosis. Blood samples are sent to the laboratory for metabolite analysis. Laboratory studies show an elevated level of ammonia in the blood. Which of the following amino acids is also most likely to be elevated in his blood?

Comment: Glutamine becomes elevated as α -keto glutarate, and glutamate scavenges ammonia from the blood. See the questions in Marks' textbook about how to differentially diagnose a UCD depending on the other metabolites observed in the blood or urine. Also, consider the appropriate interventions that should be given to a person who has hyperammonemia due to an inborn error of metabolism. Depending on the defect, the intervention might be similar or a bit different. High blood ammonia levels result in stimulation of respiratory center and hyperventilation → Respiratory alkalosis

- A. Glutamine – Correct answer –** High blood ammonia levels favor the formation of glutamine from glutamate (by glutamine synthetase)
- B. Arginine
- C. Lysine
- D. Leucine

E. Glycine

30. A 25-year-old man is brought to the emergency department because of an automobile accident. He was stranded in the mountains for 9 days. He had found shelter and water, but he did not have access to food. During this time, which of the following amino acids were catabolized for the exclusive use of ketone body synthesis?

Comment: The exclusively ketogenic amino acids are Leucine and Lysine (They form acetoacetate or acetyl CoA on catabolism). Tyrosine and isoleucine are both glucogenic and ketogenic. Glycine and alanine are only glucogenic.

A. Lysine – Correct answer

- B. Isoleucine – both glucogenic (forms propionyl CoA) and ketogenic
- C. Tyrosine – both glucogenic (forms fumarate) and ketogenic (acetoacetate)
- D. Glycine – only glucogenic
- E. Alanine – only glucogenic (Forms pyruvate)

31. A 45-year-old man comes to the physician with a 1-day history of decreased urination. He has a 3-day history of swelling of his feet. His blood pressure is 150/90mmHg. Physical examination shows bilateral pitting edema. Laboratory studies confirm acute renal disease. Which of the following parameters would be the best indicators of renal function in this patient?

Comment: Blood urea nitrogen, serum creatinine and blood uric acid levels are elevated in patients with renal disease. The liver is still functioning and forming urea, but the kidneys are not able to excrete the urea. Blood ammonia would be elevated in liver disease as the liver is unable to convert ammonia to urea.

- A. Serum ammonia – elevated in liver disease and in patients with inherited urea cycle disorders
- B. Serum albumin – levels are low in chronic liver disease. Remember, albumin is synthesized by the liver and its synthesis is reduced in patients with chronic liver disease.
- C. Blood urea nitrogen – Correct answer – Urea is made in the liver and excreted via the kidneys**
- D. Blood glucose – Are elevated in patients with diabetes mellitus
- E. Serum free fatty acids – Are elevated in patients on starvation (due to adipose tissue lipolysis), critical illness and in uncontrolled diabetics.

32. A 12-year-old girl is brought to the physician by her parents because of a 2-month history of diarrhea and skin rashes on her arms. Her 8-year-old sister has a similar history but milder clinical features. Physical examination shows a red scaly rash on her arms and neck. Neurological examination shows mild cerebellar ataxia. Dietary history shows adequate protein and niacin. Urinalysis shows elevated amounts of certain free amino acids. Which of the following amino acids is most likely found in her urine?

Comment: The patient is presenting with manifestations of niacin deficiency, pellagra. Note the neurological involvement (dementia, ataxia), diarrhea and dermatitis. Tryptophan is found in urine in Hartnup disease. The transporter for tryptophan is deficient in the GIT (reduced dietary tryptophan absorption) and in renal tubules (Increased tryptophan excretion in urine). Tryptophan is used for the formation of NAD and NADP. Patients with Hartnup disease are at risk of developing pellagra (niacin deficiency).

- A. Glutamine – Glutamine levels are elevated in blood (Ammonia levels are also elevated) in inherited urea cycle disorders
- B. Lysine – Found in urine in cystinuria (Increased losses of cystine, ornithine, arginine and lysine)
- C. Cystine - Found in urine in cystinuria (Increased losses of cystine, ornithine, arginine and lysine)
- D. Tryptophan – Correct answer – Found in urine in Hartnup disease**
- E. Glycine

33. A researcher isolates an enzyme obtained from liver biopsy specimens. It is observed that the enzyme forms free ammonia. Which of the following enzymes is most likely isolated by this researcher?

Comment: **Glutamate dehydrogenase** in the oxidative deamination direction forms free ammonia. The ammonia is used in the liver for the urea cycle. This enzyme uses NAD. **Glutaminase** can also form free ammonia. Glutaminase can release ammonia for the urea cycle in the liver. In the renal tubule, glutaminase is useful for the formation of ammonia which is lost as ammonium (NH_4^+) in the urine and is important for H^+ excretion and maintenance of acid-base balance.

- A. Alanine aminotransferase – Does NOT form free ammonia; Amino group is transferred from an amino acid (alanine) to a keto acid (alpha-ketoglutarate) to form a new keto acid (pyruvate) and a new amino acid (glutamate); Converts alanine to pyruvate and amino group is transferred to alpha-ketoglutarate that forms glutamate; Requires PLP as a coenzyme
- B. Aspartate aminotransferase – Does NOT form free ammonia; converts aspartate (amino acid) to oxaloacetate (keto acid) and amino group is transferred to alpha-ketoglutarate (Keto acid) that forms glutamate (amino acid); Requires PLP as a coenzyme
- C. Glutamate dehydrogenase – Correct answer – Glutamate is converted to alpha-ketoglutarate and free ammonia is released**
- D. Glutamine synthetase – Uses (fixes) ammonia to convert glutamate to glutamine. This is important for detoxifying ammonia in the brain.
- E. Asparagine synthetase – Uses glutamine as an amino group donor to convert aspartate to asparagine

34. A researcher is studying ammonia metabolism in the brain in experimental animals with liver disease. It is observed that ammonia is used and detoxified for the formation of an amino acid in the nervous tissue of these animals. Which of the following enzymes is most significant for performing this function in the brain?

Comment: Glutamine synthetase may be thought of as a scavenger of free ammonia in the peripheral tissues like brain. The enzyme uses free ammonia to convert glutamate to glutamine. Glutamine is a non-toxic transport form of ammonia. In liver disease, as the blood ammonia levels increase (Impaired urea cycle), plasma glutamine levels are elevated.

- A. Alanine aminotransferase – Does NOT use free ammonia; Amino group is transferred from an amino acid (alanine) to a keto acid (alpha-ketoglutarate) to form a new keto acid (pyruvate) and a new amino acid (glutamate); Converts alanine to pyruvate and amino group is transferred to alpha-ketoglutarate that forms glutamate; Requires PLP as a coenzyme

- B. Aspartate aminotransferase – Does NOT use free ammonia; converts aspartate (amino acid) to oxaloacetate (keto acid) and amino group is transferred to alpha-ketoglutarate (Keto acid) that forms glutamate (amino acid); Requires PLP as a coenzyme
- C. Glutamate dehydrogenase – Can use free ammonia in the conversion of alpha-ketoglutarate to form glutamate (direction of glutamate formation); However, glutamine synthetase reaction is the most significant to detoxify ammonia in the brain!
- D. Glutamine synthetase – Correct answer**
- E. Glutaminase – hydrolyzes glutamine to release ammonia; Important in the liver to form ammonia for the urea cycle and in the renal tubule to form ammonia for pH homeostasis.

35. A 32-year-old man comes to the physician because of a 2-month history of memory loss. He has been diagnosed with depression and has drinking excessively for over one year. His major source of calories is ethanol, and he rarely eats well-balanced healthy meals. Which of the following enzyme activities is most likely impaired in this patient?

Comment: This is a typical history of a patient with Wernicke-Korsakoff syndrome; Chronic alcoholism with thiamine deficiency (B1; TPP); Typically present with memory loss, speech changes, eye problems (nystagmus); Enzymes requiring TPP are PDH complex, alpha-ketoglutarate DH complex, and branched chain alpha-keto acid (BCAA) dehydrogenase and transketolase of pentose phosphate pathway. Transketolase activity is used to assess thiamine deficiency, because it may be done easily with a blood assay, since the PPP is active in RBC.

- A. Propionyl CoA carboxylase – requires Biotin
- B. Glutamate dehydrogenase – Converts alpha-ketoglutarate to glutamate (reversible reaction)
 - Maybe used either to synthesize glutamate (reductive)
 - Maybe used to supply NH₃ for other reactions or for elimination (oxidative deamination)
- C. Pyruvate dehydrogenase – Correct answer – requires TPP, NAD, FAD, CoA and lipoic acid**
- D. Methionine synthase – requires folate and Vitamin B12; Converts homocysteine to methionine
- E. DOPA decarboxylase – requires PLP (Used in the synthesis of Dopamine)

List some coenzymes important in amino acid metabolism and list at least one reaction requiring that coenzyme

Coenzyme	Enzyme requiring
PLP (Vitamin B6)	Transaminase, cystathionine synthase, amino acid decarboxylase
Tetrahydrobiopterin (BH ₄)	Phenylalanine hydroxylase, Tyrosine hydroxylase and tryptophan hydroxylase
TPP (Vitamin B1)	Branched chain ketoacid dehydrogenase; Pyruvate dehydrogenase; Alpha ketoglutarate dehydrogenase; Transketolase
NAD ⁺ (Niacin; Vitamin B3)	Glutamate dehydrogenase

Vitamin B ₁₂ (Cobalamin)	Methylmalonyl CoA mutase; Homocysteine methyl transferase (also requires folate)
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List specialized products of the amino acids:

Amino acid	Specialized product
Phenylalanine, Tyrosine	Dopamine, Norepinephrine, Epinephrine, Melanin, Thyroid hormones
Tryptophan	Serotonin, melatonin, NAD, NADP
Arginine	Nitric oxide, Creatine
Glutamic acid	GABA, Glutathione
Histidine	Histamine
Glycine	Glutathione, Heme

Practice MCQ for **Digestion and absorption**, **explanations**

1. A researcher is studying digestion in 4-month-old infants who are exclusively breastfed. It is observed that they use lingual lipase to degrade milk triacylglycerol (TAGs). Which of the following is most likely required for the optimal activity of this enzyme?

- A. Salivary juice pH of 7
- B. Activation of the enzyme by pepsin
- C. Anchoring to the substrate by colipase
- D. TAGs with medium-chain fatty acids
- E. TAGs with dietary essential fatty acids

2. A 46-year-old woman comes to the physician because of a 1-month history of diarrhea. Laboratory studies show a deficient release of cholecystokinin. This results in reduced activation

of a specific enzyme in her GIT. Which of the following reactions is most likely impaired in this patient?

- A. Synthesis of bile salts in the liver
- B. TAG digestion in the intestine
- C. Starch digestion in the mouth
- D. Activation of pepsinogen to pepsin
- E. Activation of trypsinogen to trypsin

3. A 32-year-old woman comes to the physician because of a 2-week history of upper abdominal pain especially after eating. Physical examination shows tenderness localized to the upper right quadrant of her abdomen. Laboratory studies show elevated serum alkaline phosphatase (ALP) levels. Ultrasound of the abdomen shows numerous stones in her gallbladder. Which of the following compounds would most likely be found in low levels in her bile?

- A. Cholesterol
- B. Glycochenodeoxycholic acid
- C. Lithocholic acid
- D. Phosphatidylethanolamine
- E. Conjugated bilirubin

4. A group of researchers are studying the effect of watching food commercials on gastric secretions. A group of volunteers are shown a food commercial video and observations are made immediately after the commercial. Which of the following is most likely observed in their stomach?

- A. Hydrochloric acid is released
- B. Amylase released from chief cells
- C. Pepsin is activated to pepsinogen
- D. Parietal cells release gastric lipase
- E. Contraction of the gallbladder is stimulated

5. A research group analyzes the structure of proteins using an enzyme for their digestion. They use purified elastase to digest the protein. Which of the following amino acids will most likely be found at the carboxyl ends of the formed protein fragments?

- A. Lysine
- B. Methionine
- C. Glycine
- D. Tyrosine
- E. Arginine

6. A 78-year-old man comes to the physician because of a 1-month history of bulky and foul-smelling stools which are difficult to flush. He has a 30-year history of alcohol abuse. Laboratory

studies show steatorrhea. Which of the following functions is most likely impaired in this patient and is only performed in the liver?

- A. Synthesis of secondary bile acids from cholesterol
- B. Synthesis of bile acids using cholesterol side chain cleavage enzyme
- C. Conjugation of bile acids with glycine or taurine
- D. Release of bile salts via CFTR into the bile ducts
- E. Formation of urobilinogen from bilirubin

7. A 56-year-old woman is brought to the physician because of a 2-hour history of severe abdominal pain. She has a history of excessive ethanol ingestion. Laboratory studies show increased serum levels of an enzyme that normally digests dietary TAGs. A diagnosis of pancreatitis is confirmed. Which of the following best describes this enzyme?

- A. Synthesized as a zymogen
- B. Acts independent of bile salts
- C. Cleaves TAG to free glycerol and three fatty acids
- D. Requires the protein colipase for activity
- E. Proteolytically activated by trypsin

8. A 42-year-old woman comes to the physician because she wants to adopt healthy lifestyle changes. Her BMI is 29 kg/m². She has read that a diet rich in high-fructose corn syrup (HFCS) can lead to health problems and is planning to reduce this dietary carbohydrate. Which of the following best describes the plasma membrane transporter required for absorption of the supplemented monosaccharide in this carbohydrate?

- A. Performs facilitated transport with the gradient
- B. Found in intestinal mucosal cells and in hepatocytes
- C. Also a high affinity glucose transporter
- D. Transports fructose into the portal vein
- E. Grouped as an ABC-transporter

9. A researcher is studying the digestion of food in adult human volunteers. They observe that the process begins in the mouth and continues in the stomach and intestines. Which of the following enzymatic reactions most likely takes place in the stomach?

- A. Lactose to galactose and glucose
- B. TAGs to monoacylglycerol and 2 long-chain fatty acids
- C. Proteins to oligopeptides and smaller peptides
- D. Nucleic acids to bases and sugars
- E. Phospholipids to fatty acids and lysophospholipids

10. A research group is studying the gastrointestinal hormones involved in the digestion of food. They observe that there is a hormone released when the proton concentration in the duodenum is increased. Which of the following best describes this hormone?

- A. Synthesized by the pancreas
- B. Activates enteropeptidase
- C. Activates parietal cells in the stomach
- D. Stimulates gallbladder contraction
- E. Stimulates release of bicarbonate from the pancreas

Key: 1 **D**, 2 **E**, 3 **C**, 4 **A**, 5 **C**, 6 **C**, 7 **D**, 8 **A**, 9 **C**, 10 **E**. **Explanations and comments for practice MCQ next page.**

Comments and explanations:

1. A researcher is studying digestion in 4-month-old infants who are exclusively **breastfed**. It is observed that they use **lingual lipase** to degrade milk triacylglycerol (TAGs). Which of the following is most likely required for the **optimal activity** of this enzyme?

- A. Salivary juice pH of 7 no: The enzyme is mainly active in the stomach at low pH.
- B. Activation of the enzyme by pepsin no: Pepsin is not involved with the activation, this is performed by substrate availability.
- C. Anchoring to the substrate by colipase no: Colipase is needed for pancreatic lipase but not for lingual lipase.
- D. **TAGs with medium-chain fatty acids – Correct answer: TAGs with medium-chain fatty acids are found in human milk and are preferred substrate for lingual and gastric lipases.**
- E. TAGs with dietary essential fatty acids no: Linoleic acid and α -linolenic acid have 18 carbons and are long-chain fatty acids. TAGs with long-chain fatty acids are not preferred substrates for lingual lipase.

2. A 46-year-old woman comes to the physician because of a 1-month history of diarrhea. Laboratory studies show a deficient release of **cholecystokinin**. This results in reduced activation of a specific enzyme in her GIT. Which of the following reactions is most likely impaired in this patient?

Comment: you have to know that **enteropeptidase** is activated by CCK

- A. Synthesis of bile salts in the liver **no**: Enteropeptidase is not needed in this process.
- B. TAG digestion in the intestine **no**: CCK leads to release of pancreatic lipase but enteropeptidase is not needed for activation of pancreatic lipase.

C. Starch digestion in the mouth **no**: Enteropeptidase acts in the duodenum and is not involved with salivary amylase reaction.

D. Activation of pepsinogen to pepsin **no**: This is an autocatalytic activation due to high proton concentration in the stomach lumen.

E. **Activation of trypsinogen to trypsin – Correct answer:** Enteropeptidase proteolytically activates trypsinogen and the shorter trypsin forms a long, narrow active pocket.

3. A 32-year-old woman comes to the physician because of a 2-week history of upper abdominal pain especially after eating. Physical examination shows **tenderness** localized to the **upper right quadrant of her abdomen**. Laboratory studies show **elevated serum alkaline phosphatase (ALP)** levels. Ultrasound of the abdomen shows numerous **stones in her gallbladder**. Which of the following compounds would most likely be found in **low levels** in her bile?

A. Cholesterol **no**: High levels of cholesterol in bile would increase risk of gallstones.

B. **Glycochenodeoxycholic acid CORRECT:** This is a conjugated bile salt and is released into the bile by the liver. Low levels of conjugated bile salts can increase risk of gall stones

C. Lithocholic acid **no**: This is a secondary bile salt. Secondary bile salts are generally lost in feces.

D. Phosphatidylinositol **no**: It is mainly phosphatidylcholine that is found in bile. Low levels of phosphatidylcholine can lead to gallstones. Phosphatidylcholine and conjugated bile salts maintain cholesterol in a soluble state (emulsifying agents) and reduce the risk of gall stones.

E. Conjugated bilirubin **no**: low levels do not lead to gallstones. Bilirubin gallstones are not as common as cholesterol gallstones.

4. A group of researchers are studying the effect of **watching food commercials** on **gastric secretions**. A group of volunteers are shown a food commercial video and observations are made immediately after the commercial. Which of the following is most likely observed in their **stomach**?

A. **Hydrochloric acid is released Correct:** HCl is released from parietal cells.

B. Amylase released from chief cells **no**: It is released by the salivary gland.

C. Pepsin is activated to pepsinogen **no**: Pepsinogen is activated to pepsin.

D. Parietal cells release gastric lipase **no**: Chief cells release gastric lipase.

E. Contraction of the gallbladder is stimulated **no**: This results from CCK action.

5. A research group analyzes the structure of proteins using an enzyme for their digestion. They use purified **elastase** to digest the protein. Which of the following **amino acids** will most likely be found at the carboxyl ends of the formed protein fragments?

- A. Lysine **no**: This is found after trypsin action.
- B. Methionine **no**: This is found after chymotrypsin action.
- C. **Glycine Correct**: Elastase has a small active site and cleaves after amino acid residues with a small side chain like glycine, alanine or serine residues. The enzyme derives its name due to the fact that it cleaves elastin which is rich in glycine.
- D. Tyrosine **no**: This is found after chymotrypsin action.
- E. Arginine **no**: This is found after trypsin action.

6. A 78-year-old man comes to the physician because of a 1-month history of bulky and foul-smelling stools which are difficult to flush. He has a 30-year history of alcohol abuse. Laboratory studies show steatorrhea. Which of the following functions is most likely **impaired** in this patient and is **only performed in the liver**?

- A. Synthesis of secondary bile acids from cholesterol **no**: Liver synthesizes primary bile acids from cholesterol. Secondary bile acids are formed by the action of intestinal bacteria on primary bile acids
- B. Synthesis of bile acids using cholesterol side chain cleavage enzyme **no**: The enzyme used for bile acid synthesis is cholesterol 7 alpha-hydroxylase.
- C. **Conjugation of bile acids with glycine or taurine Correct**: Bile salts are only formed in the liver and this process is reduced in this patient with liver disease and steatorrhea.
- D. Release of bile salts via CFTR into the bile ducts **no**: Bile salts are released from the hepatocytes into the biliary canaliculi by active transport via ABC transporters.
- E. Formation of urobilinogen from bilirubin **no**: This is performed by intestinal bacteria but not by the liver.

7. A 56-year-old woman is brought to the physician because of a 2-hour history of severe abdominal pain. She has a history of excessive ethanol ingestion. Laboratory studies show increased serum levels of an enzyme that **normally digests dietary TAGs**. A diagnosis of **pancreatitis** is confirmed. Which of the following best describes this enzyme?

Comment: You have to know that the enzyme is pancreatic lipase. Pancreatic lipase levels in blood are elevated in patients with pancreatitis. Serum amylase levels are also elevated, but amylase is required for dietary carbohydrate digestion

- A. Synthesized as a zymogen **no**: This is not needed as healthy pancreatic cells do not have high TAGs, bile salts or colipase.
- B. Acts independent of bile salts **no**: Pancreatic lipase needs bile salts for emulsification of TAGs.
- C. Cleaves TAG to free glycerol and three fatty acids **no**: It cleaves TAGs to MAG and

2 fatty acids.

D. **Requires the protein colipase for activity Correct:** Colipase is not an enzyme but a co-helper for pancreatic lipase. It anchors the enzyme to its substrate.

E. Proteolytically activated by trypsin **no:** Trypsin cleaves procolipase to colipase but does not proteolytically activate pancreatic lipase.

8. A 42-year-old woman comes to the physician because she wants to adopt healthy lifestyle changes. Her BMI is 29 kg/m². She has read that a diet rich in **high-fructose corn syrup (HFCS)** can lead to health problems and is planning to reduce this dietary carbohydrate. Which of the following best describes the plasma membrane transporter required for absorption of the supplemented monosaccharide in this carbohydrate?

Comment: you have to know that fructose uses GLUT-5 for its absorption from the intestine (Luminal surface of the intestinal mucosal cell)

A. **Performs facilitated transport with the gradient Correct:** GLUT-5 is used. GLUT-5 is a facilitated transporter and transports sugar along the gradient and does not use ATP.

B. Found in intestinal mucosal cells and in hepatocytes **no:** It is found in intestinal cells and in seminal vesicles.

C. Also a high affinity glucose transporter **no:** It is in addition a low affinity glucose transporter.

D. Transports fructose into the portal vein **no:** That is performed by GLUT-2.

E. Grouped as an ABC-transporter **no:** ABC transporters transport lipid into the extracellular space.

9. A researcher is studying the digestion of food in adult human volunteers. They observe that the process begins in the mouth and continues in the stomach and intestines. Which of the following enzymatic reactions most likely takes place in the **stomach**?

A. Lactose to galactose and glucose **no:** This is performed by lactase at the brush border of intestinal cells.

B. TAGs to monoacylglycerol and 2 long-chain fatty acids **no:** This is performed by pancreatic lipase in the small intestine.

C. **Proteins to oligopeptides and smaller peptides Correct:** This is performed in the stomach by pepsin. It will be continued by trypsin in the small intestine.

D. Nucleic acids to bases and sugars **no:** This is performed by enzymes (nucleases) in the small intestine.

E. Phospholipids to fatty acids and lysophospholipids **no:** This is performed in the small intestine by pancreatic phospholipase A₂.

10. A research group is studying the gastrointestinal hormones involved in the digestion of food. They observe that there is a **hormone** released when **the proton concentration in the duodenum is increased**. Which of the following best describes this hormone?

Comment: you have to know that the hormone is **secretin**. You also can rule out CCK as it has more than one correct statement for CCK

- A. Synthesized by the pancreas **no**: It is synthesized in cells of the duodenum.
- B. Activates enteropeptidase **no**: CCK activates enteropeptidase.
- C. Activates parietal cells in the stomach **no**: Gastrin activates release of HCl from parietal cells.
- D. Stimulates gallbladder contraction **no**: This is a function of CCK.
- E. **Stimulates release of bicarbonate from the pancreas Correct: Secretin leads to secretion of bicarbonate and water from the pancreas. This is needed to neutralize the acidic pH of chyme in the duodenum.**

Practice questions: Heme degradation and Jaundice

1. A 3-day-old boy who is prematurely born at 30 weeks of gestation has scleral icterus. A diagnosis of physiological jaundice is made. Which of the following findings is most likely in this patient?
 - A. Increased serum direct bilirubin
 - B. Increased serum indirect bilirubin
 - C. Increased serum direct and indirect bilirubin
 - D. Increased serum conjugated bilirubin
 - E. Increased activity of hepatic UDP-glucuronosyl transferase

2. A 5-day-old girl is brought to the emergency department by her parents because of a 6-hour history of poor feeding and lethargy. Physical examination shows scleral icterus and jaundice of skin and mucus membranes. Laboratory studies show total serum bilirubin is 40mg/dL (reference range: 1-2mg/dL). Liver biopsy studies shows traces (<5%) of UDP-glucuronosyl transferase activity. Which of the following is the most likely cause of this presentation?
 - A. Crigler-Najjar syndrome type I
 - B. Gilbert syndrome
 - C. Dubin Johnson syndrome
 - D. Crigler-Najjar syndrome type II
 - E. Physiological jaundice

3. A 10-year-old boy is brought to the emergency department by his parents because of acute onset of severe bone pain and abdominal pain. He had been diagnosed to have sickle cell anemia when he was 3-years old and has been managed appropriately. He says the pain started after participating in school sports activities the previous day. Which of the following sets of laboratory findings are most likely on admission?

	Serum unconjugated bilirubin	Serum conjugated bilirubin	Urine urobilinogen	Urine bilirubin
I	Increased	Increased	Normal	Present
II	Increased	Increased	Increased	Present
III	Increased	Normal	Increased	Absent
IV	Normal	Increased	Absent	Present
V	Increased	Normal	Absent	Absent

- A. I
- B. II
- C. III
- D. IV
- E. V

4. A 32-year-old woman comes to the physician because of a 3-day history of fever, loss of appetite and upper abdominal pain. Physical examination shows icterus and right upper abdominal tenderness. Laboratory studies show serum ALT is 1000 U/L, AST is 800 U/L (normal: 8-20U/L) and serum ALP level is 120 U/L (20-70U/L). Which of the following additional laboratory finding is most likely in this patient?
- A. Increased serum conjugated bilirubin only
 - B. Increased serum conjugated and unconjugated bilirubin
 - C. Increased serum unconjugated bilirubin only
 - D. Absence of bilirubin in urine
 - E. Presence of glucose in urine
5. A 68-year-old man comes to the physician because of a 1-month history of progressive unintentional weight loss and yellowish discoloration of skin. Abdominal ultrasonography shows a tumor encircling the common bile duct. Which of the following laboratory findings is most likely in this patient?
- A. Increased serum direct bilirubin
 - B. Increased serum direct and indirect bilirubin
 - C. Bilirubin absent in urine
 - D. Normal serum ALP levels
 - E. Increased urobilinogen in urine
6. A 40-year-old woman comes to the physician because of 4-month history of intermittent abdominal pain and passing of pale, bulky stools. Laboratory studies show serum ALP is 500 U/L (20-70 U/L); GGT is 470 U/L (40-50 U/L); ALT is 70 U/L and AST is 80 U/L (8-20 U/L); Amylase is 50 U/L (25-125 U/L); Urine urobilinogen is absent and fecal stercobilin is absent. Which of the following is the most likely diagnosis?
- A. Bile duct obstruction due to gall stones
 - B. Acute viral hepatitis

- C. Crigler Najjar syndrome type I
- D. Gilbert syndrome
- E. Chronic pancreatitis

Summary of inherited hyperbilirubinemias

Disorder	Age of onset	Degree of enzyme deficiency	Type of bilirubin in serum
Crigler Najjar syndrome type I			
Crigler Najjar syndrome type II			
Gilbert syndrome			
Dubin Johnson syndrome			

Answer key with brief explanations:

Answer Key:

1. B
2. A
3. C
4. B
5. A
6. A

Short explanations:

1. A 3-day-old boy who is prematurely born at 30 weeks of gestation has scleral icterus. A diagnosis of physiological jaundice is made. Which of the following findings is most likely in this patient?

Comment: In physiological jaundice, serum indirect bilirubin is elevated.

- A. Increased serum direct bilirubin – Direct bilirubin levels are increased in post-hepatic or obstructive jaundice (gall stones/ cancer of the head of pancreas)
- B. **Increased serum indirect bilirubin: Correct answer;** Preterm infants and full term infants have reduced activity of UDPglucuronyl transferase due to immaturity of the liver and as a result they have a reduced conjugation capacity by the liver. Neonates with physiological jaundice (3-10 days after birth) have increased serum unconjugated bilirubin (indirect bilirubin).

- C. Increased serum direct and indirect bilirubin – Direct and indirect bilirubin levels are elevated in hepatic jaundice (liver disease)
- D. Increased serum conjugated bilirubin – Direct (Conjugated) bilirubin levels are increased in post-hepatic or obstructive jaundice (gall stones/ cancer of the head of pancreas)
- E. Increased activity of hepatic UDP-glucuronyl transferase – In premature infants there is low activity of the glucuronyl transferase and reduced conjugating capacity of the liver

2. A 5-day-old girl is brought to the emergency department by her parents because of a 6-hour history of poor feeding and lethargy. Physical examination shows scleral icterus and jaundice of skin and mucus membranes. Laboratory studies show total serum bilirubin is 40mg/dL (reference range: 1-2mg/dL). Liver biopsy studies shows traces of UDP-glucuronyl transferase activity. Which of the following is the most likely cause of this presentation?

- A. **Crigler-Najjar syndrome type I : Correct answer: The very high serum bilirubin levels (usually >20mg/dL) with trace activity (<5%) of UDP glucuronyl transferase are diagnostic of Crigler Najjar syndrome type I. These patients do NOT improve with Phenobarbital. They require daily phototherapy and liver transplantation. Kernicterus is an important complication.**
- B. Gilbert syndrome – Typically presents in adolescents or young adults – about 50% activity of UDP-glucuronyl transferase; Patients have mild unconjugated hyperbilirubinemia (3-5mg/dL) especially following a brief illness
- C. Dubin Johnson syndrome - Typically presents in adolescents or young adults – Defect in the bilirubin ABC transporter in the liver and these patients have defective secretion of conjugated bilirubin into the bile. Conjugated bilirubin regurgitates back into the blood. These patients have mild conjugated hyperbilirubinemia and normal activity of UDP-glucuronyl transferase activity
- D. Crigler-Najjar syndrome type II – These patients have unconjugated hyperbilirubinemia (10-20mg/dL) and about 10-20% of UDP glucuronyl transferase activity. Enzyme activity is induced by Phenobarbital and there is improvement of jaundice with Phenobarbital. Kernicterus is an important complication.
- E. Physiological jaundice – Neonates with physiological jaundice have serum bilirubin levels usually <10 mg/dL. Jaundice usually resolves by the 7th to 10th day of life. Increased serum unconjugated bilirubin is because of increased RBC breakdown and due to reduced activity of UDPglucuronyl transferase after birth. Kernicterus is an important complication.

3. A 10-year-old boy is brought to the emergency department by his parents because of acute onset of severe bone pain and abdominal pain. He had been diagnosed to have sickle cell anemia when he was 3-years old and has been managed appropriately. He says the pain started after participating in school sports activities the previous day. Which of the following sets of laboratory findings are most likely on admission?

Comment: This is a patient with sickling crisis – resulting in hemolytic/ prehepatic jaundice (increased rate of RBC breakdown)

	Serum unconjugated bilirubin	Serum conjugated bilirubin	Urine urobilinogen	Urine bilirubin
I	Increased	Increased	Normal	Present

II	Increased	Increased	Increased	Present
III	Increased	Normal	Increased	Absent
IV	Normal	Increased	Absent	Present
V	Increased	Normal	Absent	Absent

- A. I – findings in hepatocellular jaundice
- B. II - findings in hepatocellular jaundice
- C. **III – Correct answer:** Serum unconjugated bilirubin is increased due to excessive RBC breakdown and the rate of heme conversion to bilirubin (Unconjugated bilirubin) is higher than the conjugation capacity of the liver. Serum conjugated bilirubin is normal as the excess conjugated bilirubin formed in the liver is excreted via the bile duct into the intestine. There is excessive formation of urobilinogen in the intestine which is found in urine. Urine bilirubin is absent as unconjugated bilirubin is not lost in urine (as it is bound to albumin)
- D. IV – Findings in obstructive/ posthepatic / cholestatic jaundice
- E. V – Urine urobilinogen is increased in hemolytic jaundice (not absent)

4. A 32-year-old woman comes to the physician because of a 3-day history of fever, loss of appetite and upper abdominal pain. Physical examination shows icterus and right upper abdominal tenderness. Laboratory studies show serum ALT is 1000 U/L, AST is 800 U/L (normal: 8-20U/L) and serum ALP level is 120 U/L (20-70U/L). Which of the following additional laboratory finding is most likely in this patient?

Comment: This is a patient with acute hepatocellular disease (acute viral hepatitis) – indicated by the very high ALT and AST levels and ALT>AST levels.

- A. Increased serum conjugated bilirubin only – Seen in patients with obstructive/ posthepatic/ cholestatic jaundice; Serum ALP levels would be very high in obstructive jaundice
- B. **Increased serum conjugated and unconjugated bilirubin – Correct answer;** Serum conjugated bilirubin is elevated because of intrahepatic cholestasis due to inflammatory process and serum unconjugated bilirubin is elevated because of reduced uptake and conjugation capacity of the liver due to the hepatocyte infection. Increased conjugated bilirubin in blood results in the appearance of bilirubin in urine and orange-yellow urine
- C. Increased serum unconjugated bilirubin only – Seen in hemolytic or prehepatic jaundice
- D. Absence of bilirubin in urine – Seen in hemolytic or prehepatic jaundice
- E. Presence of glucose in urine – This is seen in patients with type 1 and type 2 diabetes mellitus

5. A 68-year-old man comes to the physician because of a 1-month history of progressive unintentional weight loss and yellowish discoloration of skin. Abdominal ultrasonography shows a tumor encircling the common bile duct. Which of the following laboratory findings is most likely in this patient?

Comment: This is a patient with obstructive/ posthepatic jaundice

- A. **Increased serum direct bilirubin – Correct answer; Conjugated bilirubin regurgitates back into the blood and is elevated in serum. Conjugated bilirubin will be lost in urine and urine becomes orange-yellow in color. The obstruction at the common bile duct reduces the entry of conjugated bilirubin into the intestine resulting in decreased formation of urobilinogen in intestine and decreased stercobilin in feces. Also, there is reduced/ absent urobilinogen in urine (depends on the degree of obstruction)**
- B. Increased serum direct and indirect bilirubin – this is found in hepatic jaundice; Direct bilirubin is elevated due to decreased secretion into bile due to intrahepatic cholestasis; And indirect bilirubin is elevated due to decreased uptake and conjugation capacity of diseased liver
- C. Bilirubin absent in urine – this is found in patients with increased unconjugated bilirubin (hemolytic jaundice)
- D. Normal serum ALP levels – In obstructive jaundice serum ALP and GGT levels are highly elevated
- E. Increased urobilinogen in urine – Is found in hemolytic jaundice

6. A 40-year-old woman comes to the physician because of 4-month history of intermittent abdominal pain and passing of pale, bulky stools. Laboratory studies show serum ALP is 500 U/L (20-70 U/L); GGT is 470 U/L (40-50 U/L); ALT is 70 U/L and AST is 80 U/L (8-20 U/L); Amylase is 50 U/L (25-125 U/L); Urine urobilinogen is absent and fecal stercobilin is absent. Which of the following is the most likely diagnosis?

Comment: Laboratory findings (enzymes) and clinical presentation indicate obstructive jaundice

- A. **Bile duct obstruction due to gall stones – Correct answer; ALP and GGT are increased much more than ALT or AST levels. Absence of urine urobilinogen and fecal stercobilin indicate complete bile duct obstruction**
- B. Acute viral hepatitis – ALT and AST levels are very high (compared to ALP and GGT levels); Direct and indirect bilirubin fractions are elevated
- C. Crigler Najjar syndrome type I – Liver enzyme levels are usually within normal limits; Unconjugated (indirect) bilirubin levels are elevated
- D. Gilbert syndrome – Liver enzymes usually within normal limits; Unconjugated (indirect) hyperbilirubinemia
- E. Chronic pancreatitis – Usually serum amylase and lipase levels are elevated; Liver enzymes are generally normal

Summary of inherited hyperbilirubinemias

Disorder	Age of onset	Biochemical defect	Type of bilirubin in serum
Crigler Najjar syndrome type I	Very early – newborn	Complete deficiency (Less than 2% activity) of UDP glucuronyl transferase	Indirect or unconjugated; very high levels of indirect bilirubin

Crigler Najjar syndrome Type II	Milder than type I - newborn	10-20% of activity of UDP glucuronyl transferase is present	Indirect or unconjugated; respond to phenobarbital
Gilbert syndrome	Adolescence or young adults	About 50% activity of UDP glucuronyl transferase is present	Indirect or unconjugated; Mild elevations following illness
Dubin Johnson syndrome	Adolescence or young adults	Defect in the hepatocyte bilirubin ABC transporter	Direct or conjugated bilirubin is elevated

Practice MCQ for Heme synthesis and Porphyria explanations

1. A 3-year-old girl is brought to the physician by her parents because of a 2-week history of pale appearance, irritability and vomiting. The mother says that she likes to play with old toys with lead-based paint. Laboratory studies confirm lead poisoning. Which of the following enzymes is most likely inhibited in this patient?

- A. Delta-ALA synthase
- B. Delta-ALA dehydratase
- C. Hydroxymethylbilane (HMB) synthase
- D. Uroporphyrinogen-III synthase
- E. Uroporphyrinogen decarboxylase

2. A 3-month-old boy is brought to the physician because of a 3-day history of incessant crying and severe skin blisters. Physical examination shows blisters all over his body in various stages of healing. Urinalysis shows red color and elevated levels of porphyrins type I. Which of the following enzymes is most likely deficient in this patient?

- A. Delta-ALA synthase
- B. Porphobilinogen synthase
- C. Porphobilinogen deaminase
- D. Uroporphyrinogen III synthase
- E. Uroporphyrinogen decarboxylase

3. A research group is investigating the heme synthesis pathway and its precursors. They use glycine radiolabeled at the alpha-carbon atom. The synthesized heme molecule is observed to contain this label. Which of the following best approximates the number of radiolabeled carbon atoms in one molecule of heme synthesized in this experiment?

- A. 0
- B. 2
- C. 4
- D. 8
- E. 10

4. A 40-year-old woman is brought to the emergency department because of a 3-hour history of agitation and severe abdominal pain. She says she has experienced symptoms due to hormonal changes and hot flashes in the past 6 months. Urinalysis shows high levels of porphobilinogen. She is treated with pain medication and glucose infusion but her symptoms fail to improve within 36 hrs. Which of the following is the most appropriate next step of management?

- A. Blood transfusion
- B. Supplementation of vitamin B6
- C. Administration of barbiturates
- D. Intravenous administration of radical scavengers
- E. Intravenous administration of hemin

5. A research group investigates the synthesis of heme in erythroid cells and hepatocytes. It is observed that the process is different in the two tissues. Which of the following best describes the process in erythroid cells?

- A. Free heme accumulates
- B. Heme synthesis starts and ends in cytosol
- C. The regulated enzyme uses ALA
- D. ALA dehydratase needs PLP
- E. Ferrochelatase forms hemin

Key: 1 B, 2 D, 3 D, 4 E, 5 A

Explanations for practice MCQ for Heme synthesis and Porphyrin

1. A 3-year-old girl is brought to the physician by her parents because of a 2-week history of pale appearance, irritability and vomiting. The mother says that she likes to play with old toys with lead-based paint. Laboratory studies confirm lead poisoning. Which of the following enzymes is most likely inhibited in this patient?

Comment: Lead inhibits Delta-ALA dehydratase and ferrochelatase enzymes of the heme biosynthetic pathway.

- A. Delta-ALA synthase – requires PLP as a coenzyme
- B. **Delta-ALA dehydratase Correct:** this enzyme is also named porphobilinogen synthase and is inhibited by lead which interferes with the zinc cofactor and also binds to the sulfhydryl groups of the enzyme. The other enzyme of heme synthesis that is inhibited by lead is ferrochelatase
- C. Hydroxymethylbilane (HMB) synthase – Deficient in patients with acute intermittent porphyria
- D. Uroporphyrinogen-III synthase – Deficient in congenital erythropoietic porphyria
- E. Uroporphyrinogen decarboxylase – deficient in porphyria cutanea tarda

2. A 3-month-old boy is brought to the physician because of a 3-day history of incessant crying and severe skin blisters. Physical examination shows blisters all over his body in various stages of healing. Urinalysis shows red color and elevated levels of porphyrins type I. Which of the following enzymes is most likely deficient in this patient?

Comment: This is a patient with congenital erythropoietic porphyria; Note the early age of presentation and skin manifestations. They also have photophobia and increased hair growth.

- A. Delta-ALA synthase **no**: it is deficient in vitamin B6 deficiency
- B. Porphobilinogen synthase **no**: this enzyme is inhibited by lead (Delta-ALA dehydratase)
- C. Porphobilinogen deaminase **no**: this enzyme is deficient in AIP (HMB synthase)
- D. **Uroporphyrinogen III synthase Correct: this isozyme in bone marrow is deficient in Congenital Erythropoietic Porphyria which shows accumulation of red uroporphyrin-I and coproporphyrin-I in tissues, blood and urine.**
- E. Uroporphyrinogen decarboxylase **no**: it is deficient in Porphyria Cutanea Tarda and red uroporphyrin III accumulates in tissue, blood and urine.

3. A research group is investigating the heme synthesis pathway and its precursors. They use glycine radiolabeled at the alpha-carbon atom. The synthesized heme molecule is observed to contain this label. Which of the following best approximates the number of radiolabeled carbon atoms in one molecule of heme synthesized in this experiment?

- A. 0 **no**: the alpha-carbon of glycine is not decarboxylated and lost. It is the carbon of the alpha-carboxyl group that is decarboxylated by ALA synthase
- B. 2 **no**: ALA has one label from glycine and 2 labels would be found in porphobilinogen which is formed from 2 ALA
- C. 4 **no**: this is found in 2 porphobilinogens
- D. **8 - Correct: 4 porphobilinogens (containing each 2 labels) are linked by HMB synthase to form hydroxymethylbilane (HMB). Eight molecules of glycine are used for the synthesis of one heme**
- E. 10 **no**: only 4 porphobilinogens are used for heme synthesis

4. A 40-year-old woman is brought to the emergency department because of a 3-hour history of agitation and severe abdominal pain. She says she has experienced symptoms due to hormonal changes and hot flashes in the past 6 months. Urinalysis shows high levels of porphobilinogen. She is treated with pain medication and glucose infusion but her symptoms fail to improve within 36 hrs. Which of the following is the most appropriate next step of management?

Comment: The patient has acute intermittent porphyria due to deficiency of porphobilinogen deaminase (HMB synthase). Elevated levels of delta-ALA is responsible for the pain and sometimes psychiatric manifestations. Note there is NO photosensitivity and cutaneous lesions.

- A. Blood transfusion **no**: her problem is a liver enzyme deficiency which leads to AIP which occurs acutely when triggered, very likely in her case by hormonal changes
- B. Supplementation of vitamin B6 **no**: this would increase the coenzyme PLP for ALA synthase. In her case, ALA synthase should be inhibited.
- C. Administration of barbiturates **no**: this would stimulate CYP450 synthesis in her liver and can lead to life-threatening high ALA levels. ALA and porphobilinogen accumulate. More ALA is synthesized as ALA synthase is not feed-back inhibited by heme. ALA is normally not found in the blood but now at these very high levels, ALA is released by the liver into the blood and competes with GABA binding to its receptor. This reduces GABA action. Barbiturates are contraindicated (avoided) in acute intermittent porphyria
- D. Intravenous administration of radical scavengers **no**: Patients with AIP are not photosensitive. They do not have colored abnormal porphyrins in the blood which could trigger radical formation in the skin
- E. **Intravenous administration of hemin Correct**: Hemin is used for management. Hemin inhibits delta-ALA synthase in the liver and reduces accumulation of delta-ALA and porphobilinogen resulting in symptomatic improvement.

5. A research group investigates the synthesis of heme in erythroid cells and hepatocytes. It is observed that the process is different in the two tissues. Which of the following best describes the process in erythroid cells?

- A. **Free heme accumulates - Correct**: this is needed in erythroid cells and does not occur in hepatocytes.

-In erythroid cells presence of free heme stimulates protein synthesis and globin chain synthesis for hemoglobin formation. The mature RBC cannot synthesize hemoglobin (due to absence of nucleus). Hemoglobin synthesis occurs in erythroid cells and RBC precursors. The rate limiting factor for heme synthesis in erythroid cells is the availability of free iron. ALAS2 is inhibited at low intracellular iron levels in erythroid cells. It has an mRNA with a hairpin iron-response element which recognizes low iron levels.

-In hepatocytes, presence of free heme leads to inhibition of ALA synthase as accumulation of heme is not wanted as it damages the hepatocytes.

- B. Heme synthesis starts and ends in cytosol **no**: it starts and ends in mitochondria and it is the same as in hepatocytes.
- C. The regulated enzyme uses ALA **no**: The regulated enzyme (ALA synthase) does not use but forms ALA in both cells. It is the enzymes ALAS1 in liver and ALAS2 in bone marrow.
- D. ALA dehydratase needs PLP **no**: ALAS1 and ALAS2 need PLP as glycine is decarboxylated in both reactions. ALA dehydratase needs zinc in both cell types.

E. Ferrochelatase forms hemin **no**: Ferrochelatase forms heme in both erythroid cells and in hepatocytes as it incorporates ferrous iron into protoporphyrin IX.

Practice MCQ for ethanol and xenobiotic metabolism **with explanations**

1. Metabolic changes are observed in a group of 50-year-old women after a dinner and five glasses of beer. Which of the following metabolic changes will most likely occur in the liver during the night in these women?

- A. Inhibition of acetoacetate formation
- B. Inhibition of 3-hydroxybutyrate formation
- C. Inhibition of glycogen degradation
- D. Increased synthesis of triacylglycerol
- E. Increased gluconeogenesis

2. Metabolic changes are observed in a 19-year-old woman who has a dinner and with five alcoholic beverages. Which of the following hepatic enzymes will most likely lack its coenzyme during her overnight fast due to the ethanol metabolism?

- A. Alanine aminotransferase
- B. Lactate dehydrogenase
- C. Mitochondrial malate dehydrogenase
- D. Methionine synthase
- E. Glycerol kinase

3. A 28-year-old man comes to the physician for a routine blood test. He says that he generally does not drink alcohol but had many alcoholic drinks in a short time with some snacks the previous day. Which of the following findings are most likely in this patient?

- A. Increased triglyceride synthesis
- B. Very low levels of beta-hydroxybutyrate
- C. Elevated serum alkaline phosphatase (ALP) levels
- D. Low albumin levels
- E. Elevated plasma glucose levels

4. A 57-year-old man comes to the physician for a routine examination. Laboratory studies show elevated serum total cholesterol levels and he is prescribed statins. He is advised to avoid grapefruit juice when he is on the medication. Which of the following enzymes is most likely inhibited by a substance in this juice?

- A. Cytochrome P450 reductase

- B. UDP-glucuronosyl transferase
- C. Cytochrome oxidase
- D. CYP 2E1
- E. CYP 3A4

5. A 55-year-old woman is brought to the emergency department by her neighbor because of a suicide attempt. She had taken an overdose of Tylenol (acetaminophen) tablets because she wanted to end her life but then had changed her mind and called her neighbor for help. She is administered acetadote (N-acetyl cysteine). Which of the following best describes the mechanism of action of this drug?

- A. Activates formation of acetaminophen glucuronate
- B. Activates formation of acetaminophen sulfate
- C. Inhibits CYP2E1
- D. Activates glutathione S-transferase
- E. Prevents formation of NAPQI

Answer Key: 1 D, 2 B, 3 A, 4 E, 5 D

Explanations for practice MCQ for Ethanol and xenobiotic metabolism

1. Metabolic changes are observed in a group of 50-year-old women after a dinner and five glasses of beer. Which of the following metabolic changes will most likely occur in the liver during the night in these women?

- A. Inhibition of acetoacetate formation **no:** high levels of acetyl CoA and mitochondrial NADH favor ketone body formation
- B. Inhibition of 3-hydroxybutyrate formation **no:** this reaction is favored at high NADH levels in mitochondria and due to increased acetate formation
- C. Inhibition of glycogen degradation **no:** this is independent of the NADH/NAD⁺ ratio in cytosol and is stimulated by glucagon
- D. **Increased synthesis of triacylglycerol Correct! The high levels of cytosolic NADH favor formation of glycerol 3-P for TAG synthesis**
- E. Increased gluconeogenesis **no:** this pathway is reduced at high NADH/NAD⁺ ratio due to reduced usage of lactate, alanine, glycerol and glutamine provided via the blood.

2. Metabolic changes are observed in a 19-year-old woman who has a dinner and with five alcoholic beverages. Which of the following hepatic enzymes will most likely lack its coenzyme during her overnight fast due to the ethanol metabolism?

Comment: Following binge alcohol drinking, there is increased NADH/NAD⁺ ratio due to action of alcohol dehydrogenase and aldehyde dehydrogenase. This results in lactic acidosis and decreased gluconeogenesis (due to lactate and malate formation) and diversion of gluconeogenic precursors

- A. Alanine aminotransferase **no**: it uses PLP as a coenzyme
- B. **Lactate dehydrogenase Correct!** It needs NAD⁺ to form pyruvate from lactate. Ethanol metabolism leads to a high NADH level and a low NAD⁺ level in cytosol. Lactate levels increase in blood due to increased NADH/NAD⁺ ratio
- C. Mitochondrial malate dehydrogenase **no**: during gluconeogenesis this enzyme uses NADH to form malate from oxaloacetate in mitochondria. It is the cytosolic malate dehydrogenase which needs NAD⁺ and is lacking its coenzyme. The increased NADH/NAD⁺ ratio results in trapping of malate and less availability of oxaloacetate for gluconeogenesis
- D. Methionine synthase **no**: this enzyme requires methyltetrahydrofolate and vitamin B12 as coenzymes
- E. Glycerol kinase **no** this enzyme does not need a coenzyme, it uses ATP and glycerol as substrates.

3. A 28-year-old man comes to the physician for a routine blood test. He says that he generally does not drink alcohol but had many alcoholic drinks in a short time with some snacks the previous day. Which of the following findings are most likely in this patient?

- A. **Increased triglyceride synthesis Correct!** High ethanol levels lead in hepatocytes to TAG synthesis. Glycerol 3-P is formed from DHAP and also from free glycerol. The fatty acids are taken up from the blood and are activated to fatty acyl CoA and used for triglyceride synthesis.
- B. Very low levels of beta-hydroxybutyrate **no**: they would be very likely normal or elevated. Ethanol forms acetate which is used for ketone body synthesis.
- C. Elevated serum alkaline phosphatase (ALP) levels **no**: ethanol abuse can lead in the late phase of liver cirrhosis to increased ALP levels due to the irritation of the intrahepatic bile ducts. He is a healthy student and a one time high ethanol intake does not lead to increased levels of ALP.
- D. Low albumin levels **no**: this can be seen after prolonged ethanol abuse which would already indicate liver damage. He is a healthy individual who normally does not drink alcohol.
- E. Elevated plasma glucose levels **no**: His fasting blood glucose is most likely normal as his liver is healthy and glycogen degradation is not reduced. Binge alcohol consumption may reduce gluconeogenesis due to malate trapping and conversion of pyruvate to lactate (diversion of gluconeogenesis precursors), increasing risk of hypoglycemia.

4. A 57-year-old man comes to the physician for a routine examination. Laboratory studies show elevated serum total cholesterol levels and he is prescribed statins. He is advised to avoid grapefruit juice when he is on the medication. Which of the following enzymes is most likely inhibited by a substance in this juice?

- A. Cytochrome P450 reductase **no**: this is an enzyme working together with CYP450 and uses NADPH.
- B. UDP-glucuronyl transferase **no**: this enzyme is part of phase II detoxification and this enzyme is not inhibited by grapefruit juice
- C. Cytochrome oxidase **no**: This is an enzyme in the electron transport chain (Complex IV)
- D. CYP 2E1 **no**: this enzyme metabolizes mainly ethanol and is not inhibited by grapefruit juice.
- E. **CYP 3A4 Correct:** This enzyme metabolizes many drugs, including statins and this enzyme in intestinal mucosal cells is inhibited by grapefruit juice which acts via down regulation of

CYP3A4 synthesis. It is now known that the drug metabolism of at least many drugs is affected and drugs have to be tested whether their metabolism is affected by grapefruit juice.

5. A 55-year-old woman is brought to the emergency department by her neighbor because of a suicide attempt. She had taken an overdose of Tylenol (acetaminophen) tablets because she wanted to end her life but then had changed her mind and called her neighbor for help. She is administered acetadote (N-acetyl cysteine). Which of the following best describes the mechanism of action of this drug?

- A. Activates formation of acetaminophen glucuronate **no**: this is not the action of acetadote
- B. Activates formation of acetaminophen sulfate **no**: this is not the action of acetadote
- C. Inhibits CYP2E1 **no**: this enzyme is not inhibited by acetadote
- D. **Activates glutathione S-transferase Correct!** Acetadote stimulates the synthesis of GSH and activates the binding of glutathione to NAPQI. It can also bind to NAPQI instead of GSH and make the molecule more water soluble for detoxification
- E. Prevents formation of NAPQI **no**: it stimulates the detoxification of NAPQI for excretion.

Additional questions for heme synthesis and alcohol metabolism

1. A 57-year-old man comes to the physician with a 1-month history of fever and night sweats. Chest X-ray and laboratory studies confirm tuberculosis. He is prescribed a multi-drug therapy including isoniazid. A vitamin is also prescribed as a precaution to prevent the development of microcytic anemia. Function of which of the following enzymes would most likely be impaired if the vitamin is not supplemented to the patient?

- A. ALA synthase
- B. ALA dehydratase
- C. Porphobilinogen deaminase
- D. Uroporphyrinogen synthase
- E. Ferrochelatase

2. A 2-year-old boy is brought to the physician because of a 3-week history of irritability and vomiting. Physical examination shows an irritable child with pale conjunctivae and pale mucus membranes. The mother says that she found that one of his favorite play toys is painted with lead-containing paint. Which of the following enzymes of heme synthesis is most likely inhibited in this patient?

- A. Uroporphyrinogen decarboxylase
- B. Uroporphyrinogen synthase
- C. ALA synthase
- D. Porphobilinogen synthase
- E. Hydroxymethylbilane synthase

3. A 58-year-old man comes to the physician because of a 6-month history of abdominal swelling. He has a 10-year history of alcohol abuse. Physical examination shows hepatomegaly. Ultrasound examination shows fatty changes in the liver. Laboratory studies show high cytosolic concentration of a specific molecule resulting in damage of tubulins. Which of the following molecules is most likely responsible for the damage to these intracellular proteins?

- A. Acetate
- B. Ketone bodies

- C. Mitochondrial ethanol
- D. Acetaldehyde
- E. VLDL

4. A researcher investigates the hepatic ethanol metabolism with specific focus on MEOS and compares it to alcohol dehydrogenase (ADH). Which of the following best describes this system?

- A. Uses NADH as coenzyme
- B. Has higher affinity for ethanol than ADH
- C. Contains mainly CYP2E
- D. Forms acetate
- E. Found in mitochondria

Answer Key:

- 1 – A
- 2 – D
- 3 – D
- 4 – C

Short explanations for the questions:

1. A 57-year-old man is treated for tuberculosis and isoniazid is used. The physician administers at the same time a vitamin as precaution to prevent the development of microcytic anemia. Which of the following enzymes could otherwise become deficient of its coenzyme?

- A. **ALA synthase – Correct answer:** Requires PLP (vitamin B6)
- B. ALA dehydratase – Inhibited by lead
- C. Porphobilinogen deaminase
- D. Uroporphyrinogen synthase
- E. Ferrochelatase – Uses iron; Inhibited by lead

2. A 2-year-old boy is brought to the physician because of a 3-week history of irritability and vomiting. The child is irritated, had vomited and looks anemic. The mother found out that one of his favorite play toys is painted with lead-containing paint. Which of the following enzymes of heme synthesis is most likely inhibited?

- A. Uroporphyrinogen decarboxylase
- B. Uroporphyrinogen synthase
- C. ALA synthase
- D. **Porphobilinogen synthase – Correct answer:** This enzyme is also known as ALA dehydratase. Lead poisoning also inhibits ferrochelatase.
- E. Hydroxymethylbilane synthase

3. A 58-year-old man comes to the physician because of a 6-month history of abdominal swelling. He has a 10-year history of alcohol abuse. Physical examination shows hepatomegaly. Ultrasound examination shows fatty changes in the liver. Laboratory studies show high cytosolic concentration of a specific molecule resulting in

damage of tubulins. Which of the following molecules is most likely responsible for the damage to these intracellular proteins?

- A. Acetate
- B. Ketone bodies
- C. Mitochondrial ethanol
- D. **Acetaldehyde – Correct answer:** Acetaldehyde formed by the metabolism of ethanol is toxic to tubulins
- E. VLDL

4. A researcher investigates the hepatic ethanol metabolism with specific focus on MEOS and compares it to alcohol dehydrogenase (ADH). Which of the following best describes this system?

- A. Uses NADH as coenzyme
- B. Has higher affinity for ethanol than ADH
- C. **Contains mainly CYP2E – Correct answer**
- D. Forms acetate – forms acetaldehyde
- E. Found in mitochondria – It is found in the microsomes (endoplasmic reticulum)

Practice questions: Liver function tests

1. A 57-year-old woman is brought to the emergency department by her husband in an unconscious state. Her husband says that she had vomited blood 2 hours ago. She has a 3-year-history of chronic alcoholic cirrhosis with repeated episodes of ascites. Physical examination shows jaundice and ascites. Which of the following metabolic changes is most likely responsible for the unconscious state?

- A. Decrease in serum albumin
- B. Increase in blood ammonia
- C. Decreased secretion of conjugated bilirubin into biliary canaliculi
- D. Decreased synthesis of the clotting factors
- E. Increased formation of bilirubin in reticuloendothelial system

2. A 30-year-old man comes to the physician because of a 4-day history of yellowish discoloration of his eyes, abdominal pain and fever. Physical examination shows upper abdominal tenderness and a palpable liver. Laboratory studies are shown in the table. Which of the following is the most likely cause for the patient's symptoms?

	Patient	Normal ranges
Serum ALT	750	8-20 IU/L
Serum AST	500	8-20 IU/L
Serum ALP	75	20-70 IU/L
Serum GGT	80	40-60 IU/L

Serum albumin	3.2	3-5 gm/dL
Serum bilirubin	4.8	0.8-1 mg/dL
Serum direct bilirubin	2.7	0.1-0.2 mg/dL

- A. Gilbert syndrome
- B. Dubin-Johnson syndrome
- C. Acute viral hepatitis
- D. Chronic alcoholic cirrhosis
- E. Gall stone obstructing the common bile duct

3. A 45-year-old woman comes to the physician because of a 3-week history of passing pale colored and bulky stools. She also has intermittent abdominal pain usually after meals. Abdominal ultrasound examination shows gall stones in the common bile duct. Which of the following laboratory findings is most likely in this patient?

- A. Increased serum indirect bilirubin and normal direct bilirubin
- B. Increased serum ALP and normal GGT levels
- C. Increased serum GGT and normal ALP levels
- D. Increased serum ALP and GGT levels
- E. Decreased serum albumin levels

4. A 57-year-old woman is brought to the emergency department by her friend in an unconscious state. Her friend says that she had vomited blood 2 hours ago. Physical examination shows jaundice and ascites. She has a 3-year-history of chronic alcoholic cirrhosis with repeated episodes of ascites. Which of the following is most likely responsible for the increased risk of bleeding in this patient?

- A. Reduced capacity of the liver to detoxify ammonia
- B. Reduced capacity of the liver to synthesize albumin
- C. Reduced uptake and conjugation of bilirubin
- D. Reduced synthesis of the clotting factors
- E. Reduced synthesis of the conjugated bile salts

5. A 57-year-old man is brought to the emergency department by his wife in an unconscious state. His wife says that he had vomited blood 2 hours ago. Physical examination shows jaundice and ascites. Laboratory studies are shown in the table. Which of the following is the most likely diagnosis?

	Patient	Normal ranges
Serum ALT	150	8-20 IU/L
Serum AST	300	8-20 IU/L
Serum ALP	100	20-70 IU/L
Serum GGT	90	40-60 IU/L
Serum albumin	2.3	3-5 gm/dL
Serum bilirubin	3.2	0.8-1 mg/dL
Direct bilirubin	1.7	0.1-0.2 mg/dL
Serum ammonia	180	<50 mcg/L
Blood urea nitrogen	10	7-18 mg/dL

- A. Carcinoma of head of pancreas
- B. Chronic alcoholic cirrhosis
- C. Chronic renal failure
- D. Sick cell crisis
- E. Acute viral hepatitis

Laboratory findings	Acute hepatocellular damage	Cholestatic jaundice	Chronic alcoholic cirrhosis
Serum total bilirubin			
Serum conjugated bilirubin			
Serum unconjugated bilirubin			

ALT			
AST			
ALP			
GGT			
Serum albumin			

Answer Key:

1. B
2. C
3. D
4. D
5. B

Short explanations:

1. A 57-year-old woman is brought to the emergency department by her husband in an unconscious state. Her husband says that she had vomited blood 2 hours ago. She has a 3-year-history of chronic alcoholic cirrhosis with repeated episodes of ascites. Physical examination shows jaundice and ascites. Which of the following metabolic changes is most likely responsible for the unconscious state?

A. Decrease in serum albumin – is partly responsible for the ascites; Portal hypertension also contributes to development of ascites.

B. **Increase in blood ammonia – CORRECT ANSWER** – When there is massive hepatocellular damage, liver has reduced capacity for detoxifying ammonia to urea, resulting in elevated blood ammonia levels

C. Decreased secretion of conjugated bilirubin into biliary canaliculi – Is responsible for the increase in serum conjugated bilirubin

D. Decreased synthesis of the clotting factors – Is responsible for the prolonged prothrombin time (INR)

E. Increased formation of bilirubin in reticuloendothelial system – This does not occur in hepatic jaundice. The cause for the increased serum indirect bilirubin is reduced uptake and conjugation capacity of the liver

2. A 30-year-old man comes to the physician because of a 4-day history of yellowish discoloration of his eyes, abdominal pain and fever. Physical examination shows upper abdominal tenderness and a palpable liver. Laboratory studies are shown in the table. Which of the following is the most likely cause for the patient's symptoms?

	Patient	Normal ranges
Serum ALT	750	30-40 IU/L
Serum AST	500	30-40 IU/L
Serum ALP	75	40-50 IU/L
Serum GGT	80	40-60 IU/L
Serum albumin	3.2	3-5 gm/dL
Serum bilirubin	4.8	0.8-1 mg/dL
Serum direct bilirubin	2.7	0.1-0.2 mg/dL

A. Gilbert syndrome – In patients with Gilbert syndrome, there is mild increase in unconjugated bilirubin due to reduced activity of bilirubin UDP-glucuronyl transferase (50% activity). Serum enzymes are typically within normal limits

B. Dubin-Johnson syndrome – These patients have increased direct bilirubin due to an inherited defect in the bilirubin ABC-transporter; Liver enzymes are typically normal

C. **Acute viral hepatitis – CORRECT ANSWER** – Note the elevated ALT>>>AST levels; ALP and GGT increase is smaller – indicates intrahepatic cholestasis; Normal serum albumin indicates an acute disorder; Elevated serum direct and indirect bilirubin levels

D. Chronic alcoholic cirrhosis – In chronic alcoholic cirrhosis, AST:ALT ratio is 2; serum albumin levels are low; ALP and GGT elevated due to intrahepatic fibrosis; There is an increase in both direct and indirect bilirubin

E. Gall stone obstructing the common bile duct – ALP and GGT elevation is significant; ALT and AST levels may be increased slightly due to cholestasis (back pressure destroying the hepatocytes); A significant increase in direct bilirubin (due to cholestasis)

3. A 45-year-old woman comes to the physician because of a 3-week history of passing pale colored and bulky stools. She also has intermittent abdominal pain usually after meals. Abdominal ultrasound examination shows gall stones in the common bile duct. Which of the following laboratory findings is most likely in this patient?

A. Increased serum indirect bilirubin and normal direct bilirubin – This is seen in patients with prehepatic or hemolytic jaundice and in patients with Gilbert syndrome or Crigler Najjar syndrome Types-I and II

B. Increased serum ALP and normal GGT levels – Seen in rickets and osteomalacia and bone disorders due to ALP from bone

C. Increased serum GGT and normal ALP levels – Seen in patients with recent alcohol use or use of drugs which act as enzyme inducers

D. Increased serum ALP and GGT levels – CORRECT ANSWER – seen in patients with cholestasis or obstructive jaundice

E. Decreased serum albumin levels – Seen in patients with chronic liver disease

4. A 57-year-old woman is brought to the emergency department by her friend in an unconscious state. Her friend says that she had vomited blood 2 hours ago. Physical examination shows jaundice and ascites. She has a 3-year-history of chronic alcoholic cirrhosis with repeated episodes of ascites. Which of the following is most likely responsible for the increased risk of bleeding in this patient?

A. Reduced capacity of the liver to detoxify ammonia – Responsible for encephalopathy and unconsciousness in fulminant hepatic disease

B. Reduced capacity of the liver to synthesize albumin – responsible for ascites

C. Reduced uptake and conjugation of bilirubin – Responsible for the jaundice; Results in increased serum unconjugated bilirubin. Direct bilirubin is elevated due to decreased secretion into biliary tract because of the obstruction due to fibrosis

D. Reduced synthesis of the clotting factors – CORRECT ANSWER - This is responsible for the prolongation of prothrombin time in patients with chronic liver disease

E. Reduced synthesis of the conjugated bile salts – May result in fat malabsorption and steatorrhea in patients with chronic liver disease

5. A 57-year-old man is brought to the emergency department by his wife in an unconscious state. His wife says that he had vomited blood 2 hours ago. Physical examination shows jaundice and ascites. Laboratory studies are shown in the table. Which of the following is the most likely diagnosis?

	Patient	Normal ranges
Serum ALT	150	8-20 IU/L
Serum AST	300	8-20 IU/L
Serum ALP	100	20-70 IU/L
Serum GGT	90	40-60 IU/L
Serum albumin	2.3	3-5 gm/dL
Serum bilirubin	3.2	0.8-1 mg/dL
Direct bilirubin	2.7	0.1-0.2 mg/dL
Serum ammonia	180	<50 mcg/L

Blood urea nitrogen	10	7-18 mg/dL
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A. Carcinoma of head of pancreas – Direct (Conjugated) hyperbilirubinemia; ALP and GGT levels are elevated more than ALT and AST, indicating extrahepatic cholestasis; Serum albumin levels are normal

B. Chronic alcoholic cirrhosis – CORRECT ANSWER – the patient has laboratory findings of chronic liver disease; AST:ALT= 2; Serum albumin levels are low; Serum ammonia levels are elevated indicating massive hepatocellular damage (severe compromise of liver function) – responsible for the encephalopathy

C. Chronic renal failure – Patients with acute or chronic renal failure have elevated serum creatinine and blood urea nitrogen levels; Enzyme levels will be within normal limits

D. Sickle cell crisis – Indirect bilirubin is elevated; Liver enzymes are normal

E. Acute viral hepatitis – ALT levels are higher than AST; Serum albumin levels are normal

Summary of lab findings in liver disorders:

Laboratory findings	Acute hepatocellular damage	Cholestatic jaundice	Chronic alcoholic cirrhosis
Serum total bilirubin	↑↑	↑↑	↑
Serum conjugated bilirubin	↑	↑↑	↑
Serum unconjugated bilirubin	↑	N/↑	↑
ALT	↑↑↑↑	N/↑	↑
AST	↑↑↑	N/↑	↑↑ (AST: ALT>2:1)
ALP	↑	↑↑↑↑	↑
GGT	↑	↑↑↑↑	↑
Serum albumin	N	N	↓

